

Research & Development Project
B.Sc. in Digital Medical Technologies

**Predicting Infections during Hospitalization among
Postsurgical Patients**



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Date of submission: 17.09.2026

Academic Year: 2025-2026



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Project proposal form

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Hospital-acquired infections (HAIs) represent a significant threat to healthcare systems worldwide, as they are associated not only with prolonged hospital stays and extended antibiotic therapy, but also with an increased risk of morbidity and mortality. These infections are considered among the most critical complications of modern medicine and are directly influenced by factors such as the use of invasive medical devices, inappropriate antimicrobial therapy, patient age, and immunological status[1]. HAIs substantially worsen clinical outcomes and may lead to severe deterioration, including life-threatening complications.

Recent reports indicate that HAIs affect a considerable proportion of hospitalized patients. In high-income countries, approximately 7 out of every 100 hospitalized patients acquire at least one HAI, while in low- and middle-income countries this number rises to about 10 per 100 patients. Among patients who develop HAIs, roughly 10% die as a direct or indirect consequence of the infection [1]. Beyond the profound human cost, HAIs also impose a substantial economic burden. In the United States alone, they account for an estimated annual cost of \$28.4-45 billion, largely due to prolonged hospitalizations and complex treatments[1]. Notably, this estimate includes only infections caused by multidrug-resistant bacteria, which are increasingly recognized as a major contributor to the overall burden of HAIs.

Therefore, timely and accurate identification of patients at risk is essential for the development of effective infection prevention and control (IPC) strategies. HAIs can lead to clinical deterioration and increased morbidity, ultimately negatively affecting hospitalization outcomes. Furthermore, the development of an HAI often necessitates additional interventions, including prolonged or intensified antibiotic therapy, invasive diagnostic procedures, and patient isolation. These measures increase the complexity of care, place a heavy burden on medical teams, and may further elevate the risk of adverse outcomes, including mortality. Preventing and controlling HAIs remains a persistent challenge due to the wide range of risk factors, substantial inter-patient variability, and the difficulty of detecting infection development at an early stage during hospitalization.

The aim of this project is to identify the risk of hospital-acquired infections as early as possible using artificial intelligence-based models, ideally before clinical infection manifests. Early risk detection would enable timely intervention by medical teams to prevent infection onset. Ultimately, this approach seeks to improve patient care, increase the likelihood of recovery, reduce unnecessary antibiotic use, and alleviate the burden on healthcare system resources.

Project Goals:

- Develop a data-driven machine learning model to predict the risk of hospital-acquired infections during patient hospitalization.

- Enable early identification of patients at elevated risk in order to support timely and targeted preventive interventions.
- Identify key clinical and demographic variables that are strongly associated with infection risk.
- Evaluate the contribution of these variables to the prediction performance.
- Compare the performance of different predictive models and select the most accurate and reliable approach.

Expected Outcomes:

- A validated predictive model for identifying the risk of hospital-acquired infections.
- Improved understanding of infection-related risk factors based on data-driven analysis.
- Support for the development of more effective infection prevention and control strategies within the hospital setting.
- Improved patient safety and reduced rates of hospital-acquired infections.
- Enhanced, data-driven clinical decision-making and more efficient use of healthcare system resources.

The Proposed solution: methods and performance indicators

The proposed solution is the development of a machine learning-based predictive framework for early identification of patients at increased risk of developing hospital-acquired infections (HAIs) during hospitalization. The model will be constructed using structured clinical data extracted from hospital electronic health record (EHR) systems, integrating demographic, clinical, and hospitalization-related variables.

To achieve the project goals, a comprehensive data processing pipeline will be implemented. This will include data cleaning, normalization, handling of missing values using appropriate imputation techniques, feature engineering, and exploratory data analysis to characterize variable distributions and identify potential predictors of infection risk. Feature selection methods will be applied to reduce dimensionality and improve model performance and interpretability.

Multiple supervised classification algorithms will be trained and evaluated, including logistic regression as a baseline model, alongside additional machine learning classifiers such as tree-based models and ensemble methods. Model training will be conducted using stratified train-test splits, and performance robustness will be assessed through cross-validation[2].

Hyperparameter tuning will be performed using systematic search strategies to optimize predictive performance.

Model evaluation will rely on a comprehensive set of performance metrics, including accuracy, precision, recall, F1-score, and area under the receiver operating characteristic curve (ROC-AUC)[2]. Special attention will be given to recall and precision for the high-risk class, in order to balance early detection of at-risk patients with minimization of false positive predictions.

To enhance transparency and clinical relevance, explainable artificial intelligence (XAI) techniques will be incorporated, such as feature importance analysis and post-hoc explanation methods[3]. These approaches will facilitate identification of key variables driving model predictions and support clinical interpretability.

Successful completion of the project will be demonstrated by the development of a validated, robust, and interpretable predictive model with strong discriminative performance and clinically meaningful insights. The resulting framework is expected to support early intervention strategies and inform data-driven infection prevention and control practices within hospital settings.

The research and development plan:

1. Literature Review:
Conduct a broad review of current research on hospital-acquired infections, including risk factors, prevalence, and existing predictive models. This review will help guide model design and support feature selection.
2. Data Acquisition:
Obtain retrospective clinical data, including patient demographics, vital signs, laboratory results, and hospitalization information, while ensuring full compliance with privacy and ethical standards.
3. Data Preprocessing and Exploration:
Clean and preprocess the data, handle missing values and standardize variables. Perform exploratory analysis to understand patterns, detect outliers and identify variables related to infection risk.
4. Feature Engineering and Model Development:
Construct meaningful features and select the most informative predictors. Develop multiple machine-learning models, such as logistic regression and tree-based classifiers, and train them on the prepared dataset.
5. Model Evaluation and Interpretation:
Validate the models using cross-validation and test sets. Evaluate performance using metrics such as accuracy, precision, recall, F1-score, and AUC. Apply explainable AI techniques to interpret model outputs and identify key predictors with clinical relevance.
6. Results Analysis and Reporting:
Analyze the results in a clinical context and summarize insights related to early infection

detection. Document the methodology, findings and recommendations in a clear report supported by visualizations and performance summaries.

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- 3) Feng, Ting, David P. Noren, Chaitanya Kulkarni, Sara Mariani, Claire Zhao, Erina Ghosh, Dennis Swearingen, Joseph Frassica, Daniel McFarlane, and Bryan Conroy. **"Machine Learning-Based Clinical Decision Support for Infection Risk Prediction."** *Frontiers in Medicine* 10 (2023): [Frontiers | Machine learning-based clinical decision support for infection risk prediction](#)

Risk analysis table

Description of risk (Severity: Minor, Severe, Critical; Likelihood: Low/Medium/High)	Proposed risk-mitigation measures
Missing or incomplete clinical data may reduce model accuracy (Severity: Severe; Likelihood: High)	Apply data cleaning and preprocessing, handle missing values and prioritize variables with sufficient data availability.
Imbalanced data due to low infection rates (Severity: Severe; Likelihood: High)	Use appropriate evaluation metrics (Recall, F1-score, AUC) and apply class balancing techniques if needed.
Limited generalizability to other hospital populations (Severity: Moderate; Likelihood: Medium)	Validate the model using separate training and test sets and focus on widely available clinical features.



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Risk of model overfitting (Severity: Moderate; Likelihood: Medium)	Use cross-validation and compare multiple models to ensure stable performance.
Limited interpretability of model outputs for clinical decision-making (Severity: Minor; Likelihood: Medium)	Use explainable AI methods to identify key features and improve model transparency.

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Project Abstract

Hospital-acquired infections, particularly surgical site infections (SSIs), remain an important clinical challenge in surgical care. Early identification of patients at increased risk may support closer monitoring and timely clinical attention. The aim of this project was to develop and evaluate a machine-learning framework for predicting postoperative infection among surgical patients at Assuta Ramat HaHayal Medical Center.

The dataset included 35,925 surgical events, of which 141 were labelled as positive for infection. The analysis incorporated both structured clinical information and unstructured Hebrew clinical text. Structured variables included demographic characteristics, comorbidities, ASA classification, operative characteristics, as well as clinical measurements and data from the preoperative, intraoperative, and postoperative periods. The final prediction point was defined as the end of surgery. Following data cleaning, cohort definition, and feature engineering, several machine-learning models were developed and compared. The modelling process also included class-imbalance handling, feature selection, probability calibration, and decision-threshold optimisation.

The final tabular model was a Logistic Regression classifier based on 20 selected features. On the frozen test set of 9,555 surgical events, including 42 infection-positive cases, the model achieved a recall of 88.1%, precision of 4.7%, PR-AUC of 0.183, and ROC-AUC of 0.943. At the selected operating threshold, the model generated approximately 83 alerts per 1,000 procedures, with a number needed to screen of 21.4. After calibration, the Brier score was 0.0039.

For the Hebrew clinical text, TF-IDF with Logistic Regression achieved an average precision of 0.519, compared with 0.304 for TF-IDF with XGBoost and 0.183 for a locally deployed large language model on the same evaluation split. A comparison was also performed between the tabular model and a combined approach integrating the text model with the tabular model. A small improvement was observed for the combined approach during cross-validation, but this improvement was not retained on the frozen test set.

The results indicate that meaningful predictive signal can be extracted from both structured clinical data and Hebrew clinical text, while also highlighting the challenge posed by the very low prevalence of infection cases. The limited number of positive cases in the test set restricts the ability to draw definitive conclusions regarding small differences between modelling approaches, and further validation on a larger number of cases is therefore required before clinical use.

Abbreviations

Abbreviation	Meaning
AI	Artificial intelligence
ASA	American Society of Anesthesiologists physical status classification
AUC	Area under the curve
BERT	Bidirectional Encoder Representations from Transformers
BMI	Body mass index
CAUTI	Catheter-associated urinary tract infection
CDC	Centers for Disease Control and Prevention
CDSS	Clinical decision support system
CLABSI	Central line-associated bloodstream infection
CTGAN	Conditional Tabular Generative Adversarial Network
CV	Cross-validation
CVC	Central venous catheter
ECDC	European Center for Disease Prevention and Control
EDA	Exploratory data analysis
EHR / EMR	Electronic health record/electronic medical record
F1	F1-score, the harmonic mean of precision and recall
FDR	False discovery rate (Benjamini-Hochberg correction)
FN	False negative
FP	False positive
GPU	Graphics processing unit
HAI	Healthcare-associated (hospital-acquired) infection
ICD-9	International Classification of Diseases, ninth revision
ICU	Intensive care unit
IPC	Infection prevention and control

Abbreviation	Meaning
JSON	JavaScript Object Notation
L1	L1 (lasso) regularisation penalty
LLM	Large language model
LOS	Length of stay
ML	Machine learning
MRSA	Methicillin-resistant Staphylococcus aureus
NLP	Natural language processing
NNS	Number needed to screen (the inverse of precision)
OOF	Out-of-fold (cross-validated) prediction
PPV	Positive predictive value (precision)
PR-AUC / AP	Area under the precision-recall curve/average precision
ROC-AUC	Area under the receiver operating characteristic curve
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic minority over-sampling technique
SMOTENC	Synthetic Minority Over-sampling Technique for Nominal and Continuous features
SSI	Surgical site infection
TF-IDF	Term frequency-inverse document frequency
TP	True positive
UKHSA	UK Health Security Agency
VAP	Ventilator-associated pneumonia
VIF	Variance inflation factor
XAI	Explainable artificial intelligence

Introduction

Background, problem and project response

Hospital-acquired infections (HAIs) remain one of the most persistent threats to patient safety, prolonging hospital stays, causing avoidable morbidity and mortality, and consuming substantial resources, while rising antimicrobial resistance narrows the treatment options available [1]. Surgical patients are particularly exposed: surgical site infections (SSIs)- the most common hospital-acquired infection in surgical patients- develop in an estimated 2% to 5% of patients undergoing surgery [2], and in the United States alone they affect between 160,000 and 300,000 patients annually at an excess cost of US \$3.5 to \$10 billion [3]. No equivalent national figure exists for Israel, an absence discussed in detail below [6].

Detection rests on clinical assessment and manual chart abstraction by infection prevention and control (IPC) personnel. This process is supported by microbiological culture and by severity scores such as qSOFA, so that infection is generally identified only once it is already clinically apparent. Culture is adjunctive rather than diagnostic, and returns a result only after several days [4]. Manual surveillance, though the reference standard, is resource-intensive, subject to inter-observer variability, and scales with case volume, so that only a fraction of operations can be monitored [5]. In a 2022 national survey of Israeli IPC practice, surgical site infection was not covered by national surveillance at all, and no comparative national data had been collected over time [6]. Israeli IPC teams therefore operate established but under-resourced programmes [6], a setting in which automated screening could substantially ease the load on these teams by narrowing manual review to the cases that require clinical judgement.

Three families of automated alternative to manual surveillance are described in the literature. The first applies rule-based algorithms to administrative billing codes (ICD/CPT) or structured laboratory values; these reach high sensitivity but low precision, and machine-learning models applied to the same records identify infection more accurately [3]. The second applies supervised machine learning to structured tabular data alone, which predicts infection well before clinical suspicion [8], [9] but transports poorly: a systematic review of SSI prediction models has found that external validation is almost entirely absent [10], and where external validation has since been attempted, discrimination fell substantially [11]. The third adds the clinical narrative as a second modality, and evaluations against identical outcome definitions show that this combination outperforms either modality alone [3].

To address these limitations, this project develops and evaluates a multimodal Machine Learning (ML) and Natural Language Processing (NLP) framework for the early prediction of surgical-site infections (SSIs). The system operates within the hospital's existing EHR infrastructure and analyses two modalities concurrently: structured tabular data (demographics, preoperative and postoperative parameters, vital signs and laboratory results) and unstructured clinical narratives written in Hebrew. It is intended to produce an individualised risk score at

the point the surgical procedure ends, converting surveillance from a retrospective audit into a prospective decision-support function.

Any such system depends on a definition of how long after an operation an infection may still be attributed to it. The CDC's National Healthcare Safety Network defines surgical site infection by a surveillance period that begins on the day of the procedure and runs for either 30 or 90 days, the length depending on the procedure category and on the tissue depth of the infection; superficial incisional infections are monitored for 30 days for all procedure categories, while deep incisional and organ/space infections following certain procedures - including cardiac surgery, spinal fusion, and hip and knee arthroplasty - are monitored for 90 days [19]. Infections identified after the relevant period closes are not attributed to the procedure.

The operational definition used at Assuta Ramat HaHayal is the broader one. Following clinical consultation with the hospital, it was established that a postoperative infection at this medical center is any infection identified within 90 days of the index procedure, and the follow-up window applied to the dataset in this project was set to match that definition (see the Definition of the prediction target and prediction window section). A limitation follows directly from this definition. A patient whose infection is diagnosed elsewhere - at another hospital, or through their health fund - is not recorded as infected in this dataset, because the medical center has no access to those records and cannot cross-reference them against its own. The labels therefore under-count true infections rather than mislabel them: what the models learn to predict is documented infection at this institution, not every infection that occurred.

Hospitals require scalable screening tools capable of directing limited specialist personnel towards the cases most likely to represent a true infection. The dataset underlying this work was provided by Assuta Ramat HaHayal Medical Center, which is also the immediate client; the intended users and the counter-parties are identified in the Target entity, feasibility and alternatives considered section. Implementation is feasible on both technical and methodological grounds, assessed in full in the Target entity, feasibility and alternatives considered section: the system draws on retrospective EHR data already collected in routine practice and requires no additional hardware or prospective data capture, and it combines established supervised learning for tabular data with NLP techniques of documented precedent on comparable clinical corpora (Hebrew and non-Hebrew clinical text alike, reviewed in the Literature review section), including safeguards against data leakage and SHAP-based feature attribution [12]. The need addressed here is general across Israeli hospitals [6]. Assuta Ramat HaHayal is, however, a private medical center, and its case mix, admission pathways and documentation practices differ from those of the public hospital system; the approach is therefore transferable in principle, but a model developed on this cohort would require re-fitting and re-validation on local data before being deployed in a public hospital.

Theoretical background

A prediction problem of this kind lies at the intersection of statistical learning and computational linguistics. Combining a structured and an unstructured modality is generally addressed either by early fusion, which merges the modalities into one representation before learning, or by late fusion, which processes each modality independently and combines the resulting representations only at the decision stage. Late fusion is preferable when the modalities differ substantially in dimensionality, sparsity and noise, as tabular and free-text clinical data do, and when the contribution of each modality needs to be measured separately rather than assumed [3]. The present project implements a feature-level (intermediate) fusion design, described in the Clinical text processing section: the two eligible free-text fields are reduced by the best-performing text model - TF-IDF with logistic regression - to a single continuous text-derived risk score, which is merged into the tabular feature table during feature engineering (Section 5) and learned by a single model. The separate contribution of each modality is therefore established by training the same algorithm with and without the text-derived features, rather than by combining the outputs of two independent classifiers.

For the tabular component, the framework rests on tree ensembles: random forest, which averages independently grown trees over bootstrap samples, and gradient boosting (XGBoost, CatBoost), which additively fits decision trees to the residuals of the preceding ensemble; logistic regression is retained as an interpretable linear baseline. These methods suit clinical tabular data because trees capture non-linear interactions and threshold effects without explicit specification and handle mixed variable types natively. The boosting implementations can in principle consume the missing values endemic to routine records directly, learning a default branch direction at each split, but because random forest and logistic regression cannot, a single imputed feature matrix is used for all four models so that the comparison in the Structured tabular models section measures the algorithm rather than the preprocessing (Section 6); comparative studies of infection prediction consistently rank these methods at or near the top [9], [11]. At each boosting round the algorithm adds the tree that best corrects the errors the ensemble has made so far, subject to a penalty on tree complexity that limits overfitting; in this project the error term is weighted so that a missed infection costs the model more than a false alarm (Section 7).

For the narratives the project relies on natural language processing. The classical approach, TF-IDF with optional keyword rules (Formula 1), is transparent, cheap and remains a legitimate baseline [12], but it treats a document as an unordered bag of words, so "no signs of wound infection" and "signs of wound infection" yield nearly identical vectors. The weight assigned to a term is

$$\text{tf-idf}(t, d) = \text{tf}(t, d) \times \log(N / \text{df}(t))$$

Formula 1. Term weighting used by the text baselines. A term carries more weight the more often it appears in a given note, and less weight the more notes it appears in across the corpus as a whole. Wording common to every

operative note is therefore suppressed, while terms that distinguish one note from another are emphasised - which is what allows a linear classifier to be fitted directly to the notes.

Transformer models resolve the context limitation through self-attention, which conditions each token on the whole sequence, and achieve the highest reported accuracies in clinical text abstraction [7]. Language adds a further constraint. Hebrew is morphologically rich: roots inflect internally, prepositions and possessives attach as clitics, and diacritics are omitted in ordinary writing, producing high surface ambiguity and a far larger effective vocabulary than English. Sub-word tokenisation and Hebrew-specific pre-training are therefore prerequisites, and Hebrew-specific large language models, pre-trained on large general Hebrew corpora, provide that foundation [13].

Clinical applicability finally depends on Explainable Artificial Intelligence. SHAP quantifies how much each individual feature contributed to a given prediction, by comparing the prediction made with that feature against predictions made without it [12]. Applied to both laboratory values and text-derived features, it converts an opaque score into an auditable statement about which findings drove an alert, and during development an implausibly dominant feature reliably signals data leakage [12] - a diagnostic property that proved decisive in this project, as reported in the Multimodal contribution, explainability and error analysis section. Reviews of artificial intelligence in infection prevention identify explainability, alongside workflow integration and staff training, as decisive for clinical adoption [14].

Literature review

Automated infection prediction has developed in three stages. The first operated exclusively on structured data and demonstrated that early warning is attainable: Feng et al. trained gradient-boosted models on vital signs, laboratory values and demographic variables and sustained an area under the curve above 0.85 up to 48 hours before clinical suspicion, retaining comparable accuracy when restricted to a few dozen routinely collected variables [8], and Wang et al. reported similar performance in intensive care [9]. The transportability of these results, however, has not been established. In a systematic review of fifty-one surgical-site-infection prediction studies following spinal surgery, Ndjonko et al. found that artificial-intelligence models attained a C-statistic (equivalent to the area under the ROC curve) of 0.9 or above considerably more often than conventional statistical models, yet more than three-quarters reported internal cross-validation alone and none had undergone external validation [10]. Where Cui et al. did validate externally, discrimination declined from 0.986 to 0.848 [11]. Internally reported metrics should therefore be interpreted as upper bounds.

The second stage concerns the incremental value of the clinical narrative. The most direct evidence derives from studies that compare input modalities on the same surgical records: Chakraborty et al. found that the best-performing model was a neural network drawing on both structured fields and unstructured clinical notes, reaching an F1-score of 0.77 against the NHSN reference standard and 0.58 against NSQIP, and that a single model trained jointly

across both registries and all surgery types outperformed registry-specific and procedure-specific models [3]. Bonde et al. obtained higher discrimination still at note level using deep language models trained on large volumes of postoperative notes [7]. Of particular relevance here is linguistic rather than architectural evidence: Saban et al. applied natural language processing to 2.3 million Hebrew free-text records and identified clinical complications with a mean F-score of 86%, against 51% for the corresponding structured codes [15]. Although that study addressed diabetes in community records, it remains the principal demonstration that Hebrew clinical prose is machine-readable at scale.

The size of this benefit depends on how the text is represented, and that choice is the one this project had to make. Where Verberk et al. reduced the narrative to keyword counts, the gain over an existing surveillance algorithm was small: the proportion of flagged records that proved to be true infections rose by 2.6 percentage points, and the volume of records still requiring manual review fell by a further 3.6 per cent [16]. Where Flores-Balado et al. combined richer representations with gradient boosting, surveillance across four hospitals required 63.5 rather than 975 person-hours of staff time [17]. Large language models occupy an intermediate position: Casanova-Portoles et al. applied ChatGPT-4o to 1,157 colorectal procedures and detected organ/space infection with a sensitivity of 85.9% but superficial and deep incisional infection at only 21.1% and 20%, recommending its use as a screening layer rather than as a classifier [5]. These findings support the three-tier design employed here, in which TF-IDF baselines establish what frequency information alone can achieve and a Hebrew-capable contextual model is asked to recover the signal they fail to capture. Table 1 sets out the principal studies reviewed in this section on comparable axes, with the present project stated alongside them.

Table 1. Principal predictive studies reviewed, with the present project stated on the same axes. Reported performance figures are those given by the authors on their own data and are not directly comparable across cohorts of different prevalence, outcome definition and prediction point. The final column records the limitation that the present design attempts to avoid.

Study	Input	Method	Cohort	Reported performance	Limitation
Feng et al. [8]	Vitals, labs, demographics	Gradient boosting	Multi-site retrospective	AUC > 0.85 up to 48 h pre-suspicion	No external validation
Wang et al. [9]	Structured ICU record	CatBoost	ICU patients, single center	AUC 0.944, sensitivity 87.2%	ICU only; internal validation
Ndjonko et al. [10]	Systematic review, 51 studies	AI vs. statistical models	Spinal surgery SSI	C-statistic ≥ 0.9 more often for AI	> 75% internal CV only; none external
Cui et al. [11]	Structured multicenter	AI prediction model	Metastatic spinal disease	AUC 0.986 internal $\rightarrow$ 0.848 external	Substantial external drop

Study	Input	Method	Cohort	Reported performance	Limitation
Chakraborty et al. [3]	Structured + clinical text	Neural network vs. registry review	Retrospective EHR, two registries	F1 0.77 (NHSN), 0.58 (NSQIP)	English-language notes; registry-dependent
Bonde et al. [7]	Free-text EHR notes	Deep neural networks	Large postoperative note set	High note-level discrimination	Needs large annotated corpus
Saban et al. [15]	Hebrew free-text EMR	NLP vs. structured codes	2.3 million records	F-score 86% vs. 51% for codes	Community records; not infection
Verberk et al. [16]	Keyword counts from notes	Semi-automated surveillance	Colorectal surgery	+2.6 pp PPV, +3.6 pp workload cut	Sensitivity lost; weak representation
Flores-Balado et al. [17]	Structured + text features	Gradient boosting	4 Spanish hospitals	975 → 63.5 surveillance person-hours	Orthopaedic scope
Casanova-Portoles et al. [5]	EHR text	Prompted ChatGPT-4o	1,157 colorectal procedures	Organ/space 85.9%; incisional 20-21%	Screening layer only
This project	Structured + Hebrew free text	LR / RF / XGBoost / CatBoost; TF-IDF and prompted LLM	35,925 surgical events; 141 positive (0.39%)	Chapter 5	Extreme imbalance; single private center

The research gap

Three conclusions emerge from the literature reviewed above. First, structured tabular models achieve high accuracy on their development cohorts but generalise poorly and have seldom been validated externally [10], [11]. Second, the clinical narrative contains diagnostic information that structured fields omit, and the combination of the two modalities yields measurable improvement [3], [7]. Third, the magnitude of that improvement is determined by the text representation, with keyword-based methods contributing little [16] and contextual language models performing substantially better. The application of this multimodal design to Hebrew clinical documentation nevertheless remains largely unexamined: Hebrew lacks the large annotated clinical corpora available for English, and the sole large-scale demonstration to date was conducted on community rather than surgical records and did not address infection [15]. The uniform application of Explainable AI across both text-derived and tabular features similarly remains underexplored [12], [14]. Combining the two within a single evaluated system - rather than reporting a tabular model and a text model separately - is itself uncommon in the multimodal clinical-prediction literature to date, which is the specific gap this project addresses.

A separate methodological question concerns not only which variables to use but at what point in the hospitalisation a prediction should be made. Most of the SSI-specific literature reviewed above evaluates a single, retrospectively fixed prediction point, typically hospital discharge or a fixed post-operative day. Outside the SSI literature specifically, work on postoperative complications more broadly has moved towards dynamic, sequential prediction: Shickel et al. trained a multi-task deep network that updates its risk estimate continuously throughout the hospital stay as new data accrue, rather than committing to a single snapshot in time [18]. A fully dynamic, continuously updating model of this kind exceeded the scope of the present project, both in modelling complexity and in the volume of temporally resolved training data it would require from an already very small positive class. Instead, a single fixed checkpoint was adopted as a tractable approximation of the same idea, anchored at the end of the surgical procedure (see the Definition of the prediction target and prediction window section) rather than at a later post-operative point, so that the feature set never depends on monitoring data collected only because clinical suspicion had already arisen. In this specific setting that concern is concrete rather than theoretical: clinical consultation with the hospital established that post-operative data at this medical center are essentially never collected without a clinical reason, so the mere existence of such a record already indicates that a patient was considered to be at elevated risk. This is an institutional observation rather than a finding drawn from the literature, and it is developed further in the Definition of the prediction target and prediction window section.

Research questions

Based on the identified clinical needs and technological gaps in the literature, this project is guided by the following primary and secondary research questions.

Primary research question.

To what extent can a multimodal machine learning framework, combining structured tabular EHR data with unstructured narrative clinical text, predict the risk of postoperative infection at the end of the surgical procedure among surgical patients at Assuta Ramat HaHayal, compared with a single-modality tabular model?

Secondary research questions

1. Given the extreme rarity of the positive class, what combination of cohort restriction, training-set rebalancing and operating-threshold selection allows a structured tabular model to achieve a clinically useful precision-recall trade-off at the end-of-surgery prediction point?
2. To what extent, if at all, does adding NLP features derived from clinical notes improve the positive predictive value and reduce the false-positive rate of the tabular model described above, as a secondary enhancement to it?

3. Which specific features - from structured laboratory values and vital signs versus NLP-extracted clinical terms - carry the highest predictive weight in identifying early infection onset?
4. Can Explainable AI methods, specifically SHAP values, be effectively applied across this multimodal architecture to provide clinicians with transparent, actionable and trustworthy decision support?

Target entity, feasibility and alternatives considered

The target entity is a general hospital's infection prevention and control function, and the immediate client is Assuta Ramat HaHayal Medical Center, which supplied the dataset and the clinical supervision for this work. It was decided that the system addresses three groups of intended users: the managers of the pre- and postoperative departments, the clinical staff of those departments (nurses and physicians), and the hospital's IPC team. Automation of manual review has been reported to reduce surveillance effort by an order of magnitude [17]. The counter-parties are equally identifiable: clinicians who distrust systems that cannot explain themselves, and legal and regulatory functions concerned with accountability for a model-influenced decision [14].

Feasibility was considered in four respects. Technically, nothing in the pipeline is experimental: every method required - gradient-boosted trees, regularised regression, TF-IDF representation, resampling techniques, SHAP attribution and locally served LLM inference - is mature, available as open-source Python, and runs on the hardware available inside the hospital environment, with the single exception of GPU-based fine-tuning of a Hebrew Transformer, which is addressed at the end of this section. Data feasibility is the binding constraint: the extract contains 35,925 surgical events but only 141 positive cases, an imbalance ratio of approximately 225:1, and several field groups are more than 80% missing. Discrimination can still be measured, but the question that actually matters becomes precision at a clinically usable recall, and it was this that led to PR-AUC being chosen as the primary selection metric rather than accuracy or ROC-AUC (Section 8). Clinical feasibility is supported by the literature, in which comparable models have repeatedly exceeded an AUC of 0.85 at prediction points earlier than clinical suspicion [8], [9]. Organisationally the picture is the weakest of the four, and this reflects a documented national finding rather than a project-specific risk: Israeli IPC programmes are established but under-resourced, and SSI is not nationally surveyed [6]. A model that reduces review workload is therefore easier to justify organisationally than one that adds an alerting obligation.

The design adopted here - structured data and clinical text learned together - was not the only route to the same goal. Four others were considered, each of them an approach that the literature reviewed in the Literature review section has already tested, and each was set aside for a reason that literature makes explicit.

The first is rule-based surveillance over billing codes and laboratory thresholds. This is the option a hospital could put into service immediately: the logic is fully transparent and no training data are needed. Its weakness is precision. Tested against the same records, a rule-based system reached an F1 of 0.43 where a learned model reached 0.70 [3]. That gap matters here, because infection risk in this cohort is carried by many weak signals acting together, and a threshold rule cannot weigh one against another.

The second is a fixed severity score such as qSOFA. Scores of this kind have the advantage of already being familiar to clinical staff. They are, however, calculated once from admission-time values, and they were designed to flag deterioration and mortality rather than infection [4]. Neither property fits a prediction made at the end of surgery, which is the moment this project set out to serve.

The third is a fully dynamic sequential model of the kind reported by Shickel et al. [18]. Rather than producing one score at a fixed moment, a model of this kind treats the admission as a time series and re-estimates risk continuously as each new observation is recorded, so that a deteriorating patient is flagged during the postoperative course rather than at a single decision point; Shickel et al. train one such model to predict several postoperative complications at once. Methodologically this is the most capable of the four. It also requires what this dataset does not contain: timestamped longitudinal measurements, and a positive class far larger than 141 cases. It was therefore postponed rather than rejected, and it is set out as the principal direction for further development in the Recommendations for further research and development section.

The fourth is end-to-end fine-tuning of a Hebrew Transformer (AlephBERT [13]). This approach was considered during the later methodological development of the NLP component and it was attempted. It could not be completed: the hospital computing environment offered no suitable GPU, and training a model of that size against a few hundred positive cases would in any case have risked overfitting more than it promised in accuracy. Frozen inference with a locally served model was used in its place, and the change is reported as a deviation in the Comparison with the project proposal and interim report section.

Project Planning and Execution

This chapter describes what was actually built, in the order in which the pipeline executes. Figure 1 gives the overall structure; each stage corresponds to one analysis notebook and to one section below.

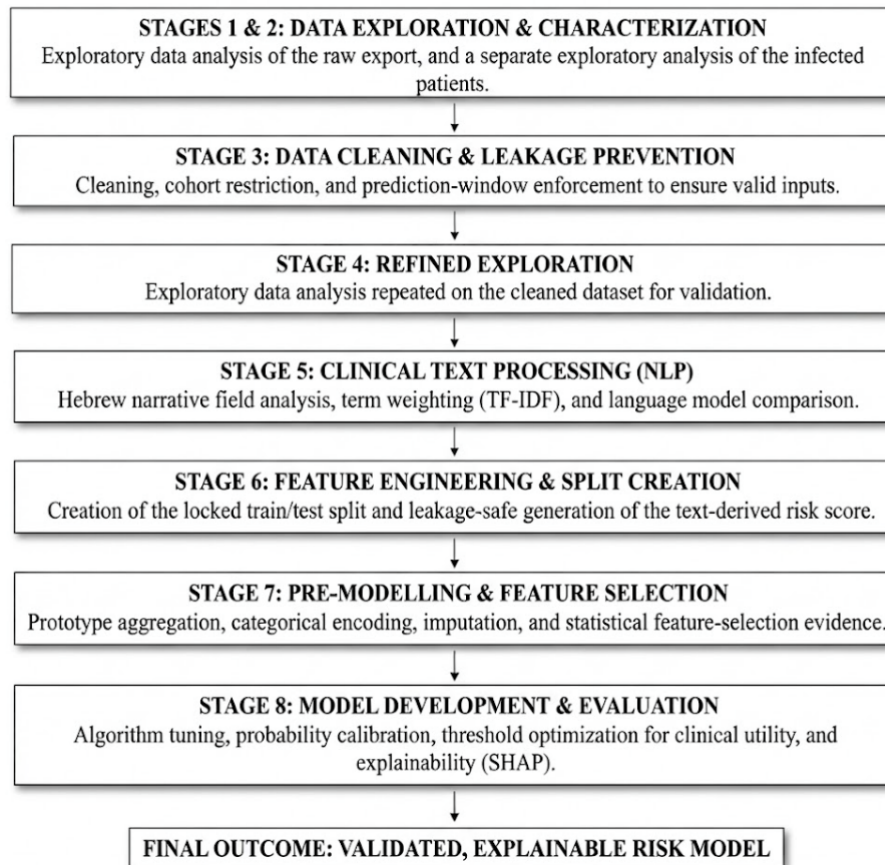


Figure 1. The project pipeline, from the raw hospital export to a calibrated and explained risk model. The eight analysis notebooks are shown in execution order.

Analysis of the literature review

Analysis of the literature identified a specific research gap. Structured fields record conclusions that have already been reached, whereas the developing clinical picture, the description of the wound and the treating physician's own suspicion are documented only as free text. Natural Language Processing is the established means of accessing that free-text content: extracting features from clinical narratives has been shown to improve model performance and reduce IPC review burden, though the size of the improvement depends strongly on the text representation used [3], [7], [16], and applying NLP to Hebrew clinical free text specifically has been shown to identify complications considerably more accurately than structured administrative codes alone [15]. On this basis, NLP is used in the present project as a secondary

enhancement to the primary tabular pipeline. The literature therefore supports both the feasibility and the necessity of a multimodal approach that combines machine learning on tabular data with large-language-model inference over Hebrew clinical free text.

This dynamic-versus-fixed trade-off, discussed in the literature review above [18], also shaped a specific design decision in the present project. Rather than assuming a prediction point from the literature, the exploratory data analysis on this cohort was used to determine, empirically, the latest point in the hospitalisation at which a risk score can still be produced without depending on data that itself reflects pre-existing clinical suspicion (see the Definition of the prediction target and prediction window section).

Beyond the treatment of free text, the reviewed literature also informs three decisions concerning the tabular component. First, studies predicting infection risk from structured hospital records consistently report that tree-based ensembles outperform linear baselines on heterogeneous electronic health record features [8], [9], [11]; boosted ensembles were therefore selected as the primary tabular model family, with a bagged ensemble retained as a stable non-linear reference point. Second, this body of work is dominated by cohorts in which infection is a rare event, and the reported performance depends heavily on how that imbalance is handled; this motivated treating cohort restriction, class rebalancing and decision-threshold selection as explicit, reported design choices rather than as defaults (Sections 6 and 8). Third, a recurring limitation identified in the systematic literature is optimistic performance produced by information that is not actually available at the intended prediction point, with leakage-controlled pipelines and post-hoc explainability recommended as the corrective [10], [12]. These three points define the methodological requirements that the pipeline described in the remainder of this chapter is designed to satisfy.

Methodology

To address the predictive challenge, several technological approaches and algorithms were evaluated for both tabular and textual data, each presenting distinct advantages and disadvantages.

Traditional statistical models (logistic regression). Highly interpretable, computationally lightweight and less prone to overfitting on small datasets; but they assume a linear relationship between features and the target, which is inadequate for the complex, non-linear physiological interactions typical of clinical EHR data.

Tree-based ensemble methods (random forest, XGBoost, CatBoost). Excellent at handling high-dimensional, non-linear data and robust to the missing values inherently present in medical records; they consistently achieve state-of-the-art performance on structured clinical tabular data [8], [9], [11], and bagged ensembles such as random forest are less sensitive to hyperparameter configuration than boosted ensembles and therefore serve as a stable non-linear

reference point. Against this, they can operate as a black box without the application of XAI techniques, and require careful hyperparameter tuning to avoid overfitting.

Traditional NLP (TF-IDF). Simple to implement and computationally efficient, transforming free text into numerical weights based on word frequency; but it operates as a bag of words, losing the semantic context, word order and nuanced linguistic dependencies of the clinical narrative.

Advanced deep-learning NLP (Transformer architectures such as BERT). Captures deep semantic meaning and bidirectional context within sentences, achieving state-of-the-art performance in clinical text abstraction; but it is computationally expensive, requires significant memory resources, and necessitates language-specific pre-training adapted to Hebrew medical terminology.

Frozen large-language-model inference (a locally hosted Llama model). Provides contextual interpretation of narrative text without any training step and without GPU resources, which makes it feasible inside a hospital computing environment; because the model is queried as a fixed function and never sees the outcome labels, it cannot overfit the few hundred positive cases in this cohort, and its structured output can be merged into the tabular table as a single feature. Against this, the score is not calibrated against the outcome of this cohort and cannot be adapted to the institution's documentation style; inference over a full cohort is slow and must be executed in batches with caching; and the internal reasoning of the model is not directly inspectable.

Based on the project goals and the data provided by Assuta Ramat HaHayal Medical Center - comprising 35,925 surgical records containing both structured preoperative and postoperative parameters and unstructured clinical summaries - a multimodal machine-learning and NLP framework was adopted.

The first component of the methodology focuses on the structured tabular data. The original proposed framework emphasised the use of gradient boosting models such as CatBoost and XGBoost, on account of their proven accuracy in forecasting clinical decline from key clinical predictors including ASA classification, glucose levels and demographic characteristics. The models selected for the final tabular pipeline are logistic regression, random forest, XGBoost and CatBoost, allowing an interpretable linear baseline and one bagged ensemble to be compared against two boosted ensembles on the same feature table and the same validation folds.

The second, critical component addresses the unstructured clinical text. Given that the clinical summaries are written in Hebrew, handling the morphological complexity of the language required a layered approach. Three text-based methodologies were implemented and evaluated, and a fourth was attempted and discontinued.

1. **TF-IDF + logistic regression (baseline 1).** The primary baseline extracts word frequencies using TF-IDF vectorisation and applies a standard logistic regression classifier. This serves as a highly interpretable, lightweight benchmark.
2. **TF-IDF + XGBoost (baseline 2).** As a secondary baseline, the same TF-IDF vectorisation was paired with an XGBoost classifier, in order to test whether a more robust, non-linear tree-based ensemble could extract better predictive value from simple frequency-based text features.
3. **Local Llama model (advanced model).** To overcome the contextual limitations of TF-IDF, a locally hosted Llama model was used as a frozen inference feature extractor, providing contextual interpretation of the clinical narratives without any fine-tuning. The model selection, prompt design, eligible input fields and execution strategy are described below.
4. **AlephBERT (attempted and discontinued).** The advanced text condition was originally specified as end-to-end fine-tuning of AlephBERT [13], a Transformer pre-trained on a large general Hebrew corpus, on the clinical summaries of the present dataset. This was attempted but could not be completed inside the hospital computing environment: without suitable GPU access the training runs were prohibitively slow, and fine-tuning a model of that size against a positive class of a few hundred cases carried a substantial risk of overfitting. The condition was therefore reallocated to the local Llama model described above, which delivers contextual interpretation of the narratives through frozen inference rather than through fine-tuning.

Ethics, data governance and working environment

The project uses retrospective, de-identified clinical records supplied by Assuta Ramat HaHayal Medical Center under the data request approved for this work, and no prospective data collection, patient contact or intervention of any kind took place. The records were released as a single export whose contents were limited in advance to the fields specified in the data request, and the study population is defined in the Cohort definition and unit of analysis section by the presence of a resolvable surgical identifier rather than by any patient-identifying attribute.

Three governance constraints shaped the technical design directly. First, all analysis was required to run inside the medical center's secure computing environment without internet access, so every model, dictionary and stop-word resource had to be held locally and the relevant libraries configured for fully offline operation; no component of the pipeline may attempt to reach an external model repository at run time. This is the reason the contextual text condition uses a locally served model rather than a commercial API, and it is a governance requirement rather than a performance preference. Second, no patient-level data leaves that environment: export cells in the notebooks are disabled by default, and only aggregate tables and figures were extracted for use in this report. Third, the surgical identifier case_number is

retained throughout as the sole linkage key between the tabular pipeline, the text pipeline and the final predictions, so that no re-identifying attribute is required to align the modalities.

Two ethical considerations concern the intended use rather than the data. A model of this kind is a screening aid and not a diagnostic device: its output is a ranked review priority for the IPC team, it is never a treatment decision, and clinical oversight remains in the decision loop at every point [14]. And because the model is trained on documentation produced by one private medical center, its errors will reflect that center's documentation practices; the subgroup analysis described in the Evaluation, threshold tuning, calibration and explainability section is included precisely so that systematically worse performance in an identifiable patient group would be visible rather than hidden inside an aggregate metric.

Cohort definition and unit of analysis

The unit of analysis in this project is the surgical event rather than the patient. Every record in the dataset corresponds to one procedure and is identified throughout the pipeline by `case_number`, which is preserved unchanged from cleaning to final prediction. Fixing the unit of analysis explicitly is a precondition for the multimodal design: the tabular features, the features derived from the clinical narratives and the model output must all be attributable to the same procedure, and any ambiguity in the identifier would silently misalign them.

The cohort consists of the surgical events supplied by Assuta Ramat HaHayal Medical Center - 35,925 procedures as received, of which 141 are labelled positive for infection, a prevalence of approximately 0.39%. Records were included on the basis of their presence in that export together with a resolvable `case_number` and the timestamps required to establish the chronology of admission, surgery and discharge; where the timestamps were missing or internally inconsistent, the affected time-derived variables were left missing rather than the record being discarded, so that the cohort is not narrowed by data-quality artefacts.

A first cohort restriction is applied at the beginning of the cleaning stage rather than at modelling time, because it defines the study population itself and not the treatment of individual values. Two criteria were used. Surgical events of patients under the age of 28 were removed, since the paediatric and young-adult population is clinically distinct and is represented too sparsely to support a separate model. In addition, four activity clusters in which no infection was observed anywhere in the export were removed as a frozen exclusion list: ear, nose and throat surgery (433 events), neurosurgery (44), plastic surgery (9) and radiosurgery (2). Together the two criteria removed 4,076 surgical events, reducing the cohort from 35,925 to 31,849 events while leaving all 141 positive cases in place, so that prevalence rises from 0.392% to 0.443%. The restriction is deliberately executed before the train/test split, because it is a statement about which surgical events the model is declared to address at all, and applying it after the split would make the two partitions describe different populations. The exclusion list is fixed in code rather than recomputed, so that the same population is reproduced on every run.

One category of record required an explicit clinical decision: death during admission. A single such case occurs in this cohort. Following clinical consultation with the hospital, it was decided that this patient should remain in the study cohort and be treated as an ordinary case rather than excluded or handled as a separate competing-risk category; the case is nonetheless tracked individually through the pipeline by case_number so that its retention can be verified after every cohort-affecting step. Whether the cohort should be narrowed further before any balancing technique is applied is addressed separately, as a population-definition decision distinct from cleaning, in the Pre-modelling, class imbalance and data partitioning section.

Definition of the prediction target and prediction window

The dataset does not contain a single ready-made outcome column. Following the labelling convention used by the hospital's own clinical systems, infection is recorded across two separate fields - one holding a general infection diagnosis and one holding a clinical surgical-site-infection diagnosis - and inspection showed that these are not populated consistently: there are records in which the first field is empty and the second is not, and records in which the reverse holds. A surgical event was therefore defined as a positive case if at least one of the two fields contains a non-empty value, and as a negative case if both are empty. This rule reproduces the distinction the source system itself already makes between infected and non-infected cases rather than introducing a new boundary chosen independently by the project team. Although the first of the two fields is a general infection-diagnosis field and is not restricted to surgical site infection by its name, inspection of its populated values showed that every recorded entry in this cohort documents a wound infection. The second field records the location of the surgical site infection, which confirms that what the two fields capture together is surgical site infection specifically and not infection of some other kind. The composite label is therefore a surgical site infection label, and the outcome is referred to as SSI throughout this report. The infection site was retained as a separate descriptive variable but is not used as a model input.

A second and equally consequential decision concerns the prediction window: the point in time, relative to surgery, at which the model is required to produce a risk score. The window is not a presentational detail, because it determines which variables are admissible at all - every field recorded after the cut-off is unusable for training and for inference alike, as discussed in the Data cleaning and prevention of data leakage section. Rather than assuming a cut-off from the literature, the window was fixed only after the exploratory data analysis reported in the Exploratory data analysis section, and its final specification reflects those findings directly. That analysis mapped both the chronological dynamics of infection-related documentation and the availability of information at each point along the care pathway, and identified three distinct time zones in which clinically meaningful information accumulates: the pre-operative phase, the early post-operative phase and the late post-operative phase. The window specified below was selected from these empirically observed zones, so that it reflects the information structure of the data itself rather than an assumption made in advance. The order of presentation in this

chapter is thematic rather than chronological: the target and window are defined here because Sections 1 to 3 refer to them throughout, but they were determined only after the exploratory work those sections describe.

Three separate checkpoints - pre-operative, 24-hour post-operative and 48-hour post-operative - were considered in parallel as an initial intention, formed while the pipeline was being developed, so that the trade-off between how early a warning is issued and how much data supports it could be quantified empirically. The EDA quantified exactly that trade-off, and in doing so identified a problem with the two post-operative checkpoints rather than only their relative merits. Cross-referencing the timing of every candidate variable against the moment of surgery showed that variables recorded in an extended post-operative window - antibiotic administration after the immediate perioperative period, blood and wound cultures, readmission fields - are populated overwhelmingly because a clinician already suspects infection, not independently of it: fewer than half of one percent of records carry a blood or wound culture at all, and where one exists it is because someone already looked. A 24- or 48-hour window would therefore not describe risk factors observed before the fact; it would selectively admit exactly the patients for whom clinical suspicion already existed, which is target leakage in substance even in fields that do not name the outcome directly. This was confirmed in clinical consultation with the hospital, where it was noted that data collected in the hours after surgery is essentially never collected without a reason - a clinician who orders an additional test in that window is already reacting to something. The three-checkpoint design therefore remained an initial idea only; the window actually implemented, described below, departs from it.

The prediction window actually implemented is therefore a single checkpoint, not three: every variable available up to and including the end of the surgical procedure itself - demographics, comorbidities, ASA classification, the planned procedure, operative duration, intra-operative vital signs and the immediate post-operative surgical note written by the operating team - is admissible, extended by a short, explicitly bounded tolerance of a few hours to accommodate ordinary documentation lag rather than to admit a second monitoring period. Any value timestamped beyond that tolerance is treated as unavailable at prediction time and excluded; where the great majority of a variable's values fall outside the tolerance for a given source, the variable is dropped outright rather than partially retained (Section 3). This is a narrower window than either post-operative option originally proposed - in this dataset, only 7-8% of first laboratory results are available before surgery, and most of the remainder is drawn at a median of 14-16 hours afterwards, so a strict end-of-surgery cut-off excludes those laboratory results by design. This is not a cost of the design but its purpose: clinical consultation with the hospital established explicitly that a laboratory test drawn in the hours after surgery is itself evidence of an existing clinical suspicion, so admitting such values would import the outcome into the feature set rather than predict it.

Even fixed this early, the window remains well ahead of the point at which infection becomes clinically apparent, so it represents a genuine early-warning opportunity rather than a retrospective description of an established infection. Both the outcome label and the window specification are fixed before any model is trained, so that every bivariate test reported in the Bivariate and multivariate analysis section and every feature admitted to the models is evaluated against the same, unambiguous reference point.

The working process

The stages described in this section form the working process actually executed, in the order in which it was run: from the exploratory analysis of the raw export, through statistical screening, cleaning and leakage control, text processing and feature construction, to partitioning, model training and evaluation.

1. Exploratory data analysis

Before any modelling was attempted, a dedicated exploratory data analysis notebook was written in Python (pandas, NumPy, SciPy, Matplotlib and Seaborn). The notebook was largely structured as reusable functions rather than as a linear script, so that the identical set of checks could be executed twice: once on the raw dataset as received, and once on the cleaned dataset after preprocessing.

The first step was semantic typing of the dataset. Because column names in the source file were inconsistent in spacing and formatting, all names were normalised before use. Each column was then assigned a semantic type - datetime, numeric, categorical, boolean or text - through a manually reviewed mapping file rather than by relying on the data types inferred automatically by pandas, since clinical exports frequently store numeric measurements and boolean flags as free text. A validation step confirmed that every column in the dataset appeared exactly once in the mapping, that no column had been assigned two conflicting types, and that no mapped column was absent from the data. This mapping, and the exploratory analysis it supports more generally, is a momentary, analysis-time construct: it determines how a column is read and summarised at the point the notebook runs, and does not itself alter the underlying data - any correction to the data itself is a cleaning decision, made explicitly and separately in the Data cleaning and prevention of data leakage section.

Data quality was then assessed at dataset level: total rows and columns, the number of missing cells, and the number of fully duplicated rows. The single most consequential property of the dataset is visible immediately in the outcome variable itself (Figure 2), and missingness was quantified per column and visualised by grouping features into clinical domains (Figure 3), as well as by generating targeted missingness heat maps for these clinical sub-groups. This targeted approach was used to identify columns whose values are absent together rather than independently. The resulting pattern is informative rather than merely inconvenient: the field groups most strongly associated with the outcome are precisely those that are populated only

once a clinician has already acted on a suspicion, which is the observation that fixed the prediction window in the Definition of the prediction target and prediction window section.

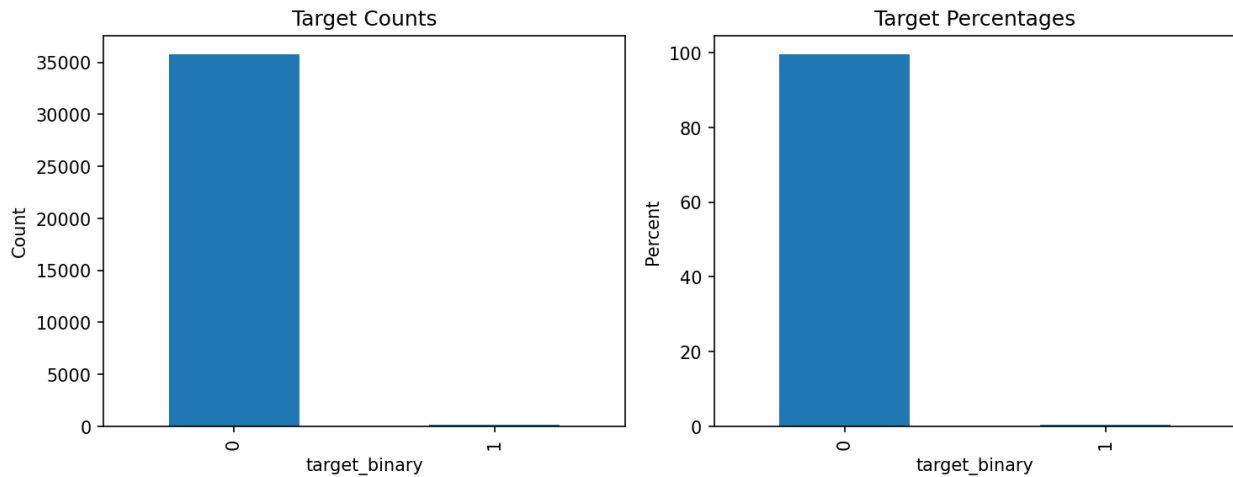


Figure 2. Distribution of the infection outcome in the raw export. Counts (left) and percentages (right) of the binary target across 35,925 surgical events. The positive class comprises 141 events, approximately 0.39% of the cohort, giving an imbalance ratio of roughly 225 negative cases to each positive case.

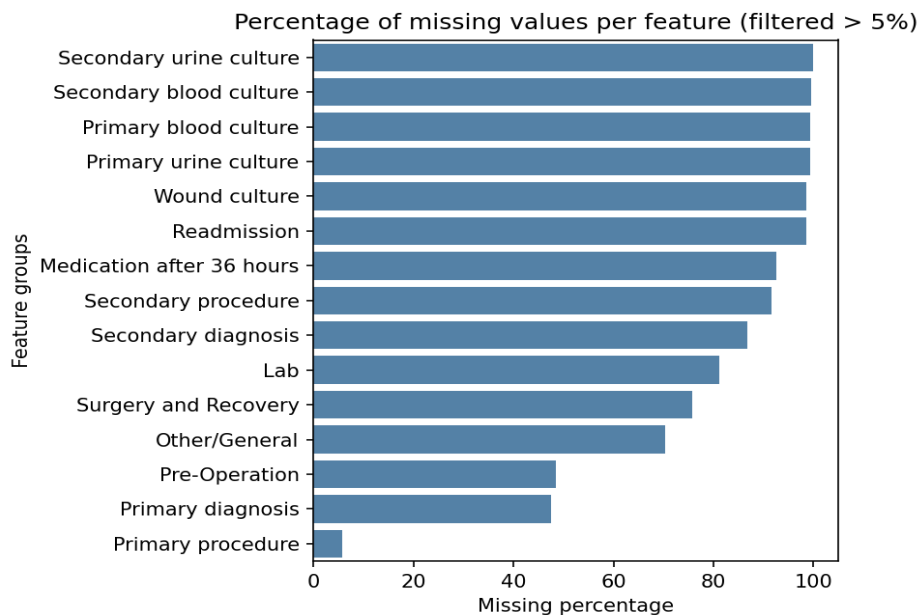


Figure 3. Missingness by clinical feature group in the raw export (groups above 5% missing). Culture groups, readmission and late medication fields are missing for almost the entire cohort, while primary procedure fields are almost complete.

Univariate analysis followed. For every numeric variable the notebook produced descriptive statistics extended with the 1st, 5th, 95th and 99th percentiles together with skewness and kurtosis, since clinical measurements are rarely normally distributed and the degree of departure from normality determines which statistical test is appropriate later. Histograms and

boxplots were generated for each variable, and statistical outliers were flagged using the interquartile-range rule.

Categorical and boolean variables were summarised by cardinality, missingness and dominant categories. Visualisations were tailored to data structure to ensure readability.

Datetime variables required separate treatment. Each was parsed with day-first interpretation, tracking conversion failures to distinguish them from inherent missing data. Temporal distributions by month, weekday and hour were plotted for the four central timeline variables.

To prevent data leakage and support future feature engineering, the analysis emphasised chronological plausibility. Clinical durations such as ICU stays were calculated to flag negative values or events occurring outside the hospitalisation window. Additionally, the availability of laboratory results, cultures and clinical events was strictly quantified relative to the surgery time - before, within 24 hours, or after. Finally, a sanity check validated the provided length of stay against a manually recalculated admission-to-discharge duration.

Textual variables underwent a specialised review detailing missingness rates, unique value counts and median text lengths. To handle the diverse nature of clinical text, these features were stratified into three processing pathways. Standard text columns were visualised using top-frequency bar charts. High-cardinality, short-text fields such as regular medications were earmarked for lightweight feature engineering, such as keyword flagging. Conversely, long-form clinical narratives such as surgical descriptions were explicitly deferred to the dedicated NLP pipeline of the Clinical text processing section.

2. Bivariate and multivariate analysis

Bivariate analysis evaluated the relationship between each predictor and the postoperative infection target. When running hundreds of statistical tests simultaneously, the chance of finding a false positive by pure luck increases. To prevent this, a strict correction factor (Benjamini-Hochberg false discovery rate) was applied to all results, ensuring that only reliable associations were highlighted.

Rather than applying a single rigid statistical test to all columns, a dynamic pipeline was implemented to select the appropriate mathematical tool based on the data's characteristics. For numeric variables the pipeline checked whether the measurements followed a normal distribution (Shapiro-Wilk test), applying standard tests (Welch's t-test) for normally distributed data and robust alternatives (Mann-Whitney U) for skewed data; alongside the p-value, the system calculated the effect size to show not just whether a difference exists but how large and clinically meaningful it is. For categorical variables, extremely rare clinical categories were grouped into an "Other" bucket to ensure mathematical stability, after which the association with the infection target was tested (chi-square or Fisher's exact test) and the strength of that link measured (Cramér's V). For datetime variables, raw timestamps were converted into clinically interpretable features - such as the month, the day of the week, or the

hours elapsed from surgery to a laboratory test - and these derived features were then processed through the numeric or categorical pipelines described above.

Significant relationships were visualised using boxplots for continuous variables and stacked bar charts for categorical distributions. Beyond these standard features, the bivariate analysis incorporated specialised checks. Free-text columns were evaluated by extracting the most frequent terms across the entire dataset to prevent outcome-based selection bias, calculating the specific infection rate for each term. Missingness was explicitly tested as an independent categorical variable to determine whether the mere absence of documentation carried a predictive clinical signal. Finally, a dedicated department-level analysis was conducted to summarise surgical volume and visualise localised infection rates.

Multivariate analysis mapped the dataset's internal structure to address multicollinearity and rigorously screen for potential data leakage. To identify feature redundancy, correlation matrices (Spearman and Pearson for numeric data, Cramér's V for categorical data) were generated, explicitly flagging highly associated pairs to inform the upcoming feature-selection phase. Simultaneously, a dedicated review pipeline flagged variables for zero variance or potential target leakage based on predefined clinical keywords such as "discharge" or "culture_result". Finally, after evaluating all features against the target, a pairplot was generated dynamically; to maintain visual readability as an exploratory visualisation, it automatically selected only the top six numeric features with the strongest target correlation, while strictly excluding any variables flagged for potential leakage.

3. Data cleaning and prevention of data leakage

The purpose of the cleaning stage was to ensure that every variable entering the pipeline carries the meaning it is assumed to carry. In a clinical export, a date read in the wrong format, an identifier treated as a continuous number or a physiologically impossible measurement can all create artificial patterns that a model will learn and exploit. Cleaning was therefore treated as a correctness requirement rather than as an optimisation step.

Each column was first assigned an explicit semantic type - numeric, datetime, categorical, text or boolean - through a manually reviewed mapping, because hospital exports frequently store measurements and flags as free text. Date conversion was made deliberately robust, since the same export may contain Excel serial numbers, integers and textual dates in the same column.

The cleaning notebook deliberately separates findings from changes. Suspicious values, near-duplicate categories, high missingness and implausible dates are first surfaced in audit tables; only rules that were explicitly reviewed and approved are then allowed to modify the working table. This conservative design reflects a property of clinical data: a value that looks technically anomalous is frequently a genuine clinical finding. Where a measurement fell outside a range agreed to be physiologically impossible it was replaced with NaN rather than clipped to the boundary of that range, because clipping would fabricate a measurement that was never taken, whereas NaN preserves the fact that the true value is unknown. For the same reason no

statistical imputation was performed at this stage; missing values were mapped and described, because a missing laboratory result may mean that the test was not indicated, that it was not performed or that the value was lost in transfer, and these are not interchangeable.

Several concrete decisions followed from this design. The recorded duration of surgery was missing for roughly 78% of the cohort, and the previous fallback of assuming a zero-minute duration effectively equated the start of surgery with its end; the end of surgery is therefore derived from the transfer time out of the surgical bed whenever the recorded duration is unavailable, which materially affects the end-of-surgery cut-off used throughout the project. Records in which the activity date preceded the admission date were handled according to the size of the discrepancy: gaps of up to 24 hours were normalised to calendar-date level, while larger gaps were treated as genuine data-quality errors and both values were set to missing rather than guessed. Boolean variables stored with textual or otherwise inconsistent representations were standardised to strict 0/1 encoding.

Columns carrying no information were removed rather than retained for completeness. Two comorbidity indicators held a single value across the entire cohort, a direct consequence of the age-based cohort restriction described in the Cohort definition and unit of analysis section, and were dropped. A third single-valued field, recording that a surgical drain had been left in place, was retained instead of dropped: it is populated for a clinically meaningful number of records, and it was converted into a binary indicator in which a populated value denotes the event and a missing value denotes its absence.

A group of columns originating from a secondary export, populated for only about a quarter of the cohort, was assessed for redundancy against the corresponding fields of the main medical record rather than removed on the basis of missingness alone. One of them duplicated an existing preoperative ASA field and was dropped outright. For the procedure fields, the free-text descriptions were first compared on matched samples and then their ICD-9 codes were compared directly, with 93% exact agreement; review of the disagreements showed that they described different steps of the same multi-step operative session rather than contradictory procedures. The duplicate procedure descriptions and codes were therefore dropped, but only after any gap in the retained ICD-9 field had first been filled from them, so that consolidation did not discard information.

Paired diagnosis and procedure fields were reconciled before any missing-value rule was applied, using an offline ICD-9 dictionary held locally so that no external service was required. Where text and code were both present the pair was audited only; where one was present and the other missing, the gap was filled only when the dictionary contained an exact, approved match; where both were missing the pair was left missing for the approved missingness rules; and ambiguous text was never converted into a guessed code but referred to manual clinical review. This prevents the common failure in which a procedure that exists but is missing its code is silently recorded as no procedure at all.

Near-duplicate categories were reviewed through a textual similarity sheet, which was treated as a review aid rather than as an automatic merging rule, since a high similarity score frequently reflects a shared prefix or a shared modality term rather than shared clinical meaning. Only two kinds of change were made: non-informative repeated prefixes were stripped, so that ward and unit identifiers were reduced to the identifying number and a repeated site suffix was removed from department names; and values were merged only where their clinical meaning at the prediction point was confirmed to be identical, as for the two nasal-specimen values that both denote a specimen taken with the result not yet available. Several column groups that the similarity sheet had flagged were reviewed and then left unchanged, because the resemblance between their values proved to be textual rather than clinical: catheter sizes, admission years and weekdays, ASA values, surgical specialties, imaging examinations and antibiotic names.

Missing values were classified before they were treated. Following clinical consultation with the hospital, columns were divided into those in which absence is clinically legitimate and those in which absence indicates a data-entry error. For the first group one of three explicit strategies was assigned per column: text and categorical fields were filled with an explicit unknown category; numeric and binary fields in which zero is already a valid value were left missing and accompanied by a separate indicator recording whether the measurement had been taken at all; and categorical fields in which absence clearly means that an intervention did not occur were filled with an explicit descriptive value. For the second group values were left missing and deferred to statistical imputation after the train/test split, so that no distribution-dependent estimate is fitted on data the model will later be tested on. The outcome fields were never modified by this process.

Throughout the stage, `case_number` was preserved as the immutable identifier of the surgical event. This is what allows the tabular pipeline, the text pipeline, the train/test partition and the final predictions to refer to the same unit of analysis without ambiguity.

Guarding against data leakage was treated as a first-class requirement rather than as a final check. Because the dataset is a retrospective export of complete hospitalisation records, it necessarily contains fields that are populated only after an infection has been identified, and a model trained on those fields would appear highly accurate while being clinically worthless. Three measures were applied. First, the columns from which the outcome label is derived were excluded from every feature list at the point where the lists were constructed, so that they could not re-enter the analysis by accident at a later stage. Second, an automated audit classified every remaining column as permitted, as a leakage source or as requiring a temporal availability check, flagging column names containing any of a predefined set of keywords - discharge, readmission, infection diagnosis, SSI diagnosis, culture result, organism detected, wound culture, after 36 hours, within 100 days, post surgery, last value and last date. Flagged columns were then reviewed manually rather than deleted automatically, since a keyword match is an indication and not a verdict.

The decisive question in this audit was not whether a variable is associated with infection but whether it would have been available at the moment the prediction is required. A microbiological culture, for example, typically returns a result 48 hours or more after the sample is taken; it is an excellent statistical predictor and a useless clinical one, because by the time it exists the infection is already known. Such variables were mapped and excluded from every feature set, so that the system predicts risk rather than describing an established diagnosis.

Two mechanisms implement the audit. An unconditional drop list removes columns that are post-window by definition - microbiological cultures and their results, antibiotics and other medications administered more than 36 hours after admission, readmission fields, late laboratory panels and the post-operative assessment. An empirical timing gate then handles the column groups whose timing varies from patient to patient. For each group the recorded timestamp is compared against the window on a row-by-row basis: if at least 70% of the documented values fall inside the window the group is retained and only the out-of-window values are set to missing, and otherwise the whole group is dropped. Thirteen groups were gated and three passed: the recovery-bed group with 98.5% of documented values inside the window (28,410 of 28,837), central venous catheters with 97.8% (2,187 of 2,236) and urinary catheters with 86.1% (4,294 of 4,987). In each of the three the group was retained and the out-of-window rows were nulled, together with their companion columns. Ten groups failed and were dropped in full. Representative examples are catheter end dates, with 2.1% of documented values inside the window, total parenteral nutrition at 5.2% and C-reactive protein at 8.0%; the remaining failures were the other first-laboratory groups, none of which reached the 70% threshold. The consequence is the single most important property of the final feature set and is referred to throughout the results: no laboratory value of any kind survives the prediction window, so the models are trained on demographic, comorbidity, procedural, device and intra-operative information only.

One family of fields required an explicit clinical decision rather than a rule. The admission-screening fields - the nasal-specimen record, the MRSA screening result, the detected-pathogen field and its specimen-source field - are microbiological in nature, which makes them superficially indistinguishable from the culture fields on the unconditional drop list. They are not, however, post-operative: at this medical center they record the screening performed on admission, before the procedure. They were therefore retained deliberately, as legitimate pre-operative predictors, and this decision is recorded here because it is the reason these fields appear among the most influential features of the final model in the Multimodal contribution, explainability and error analysis section rather than being treated there as residual leakage.

The quantitative effect of the stage is as follows. The export entered cleaning with 244 columns after type conversion and left the first phase with 93. The reduction is accounted for by the 118 columns of the leakage drop list, the ten column groups that failed the timing gate, two zero-variance comorbidity indicators and the redundant duplicates of the secondary export. A total

of 2,781 physiologically impossible values across twelve configured columns were set to missing rather than corrected. Chronologically impossible admission and surgery pairs were resolved by discrepancy size: 28 rows were normalised to calendar-date level and 18 rows had both timestamps set to missing. Gaps in the preoperative ASA field were filled from the corresponding field of the secondary export in 1,958 cases, and gaps in the primary procedure field were filled from the redundant activity and procedure-code fields in 1,942 and 1,166 cases respectively, before those redundant fields were dropped. Cleaning also added a small number of columns rather than only removing them. The boundary applied throughout is whether a transformation estimates anything from the data: these derivations are deterministic and clinically defined, so they may be applied to the whole export before the train/test split, whereas representational choices that depend on a learned parameter belong to feature engineering (Section 5) and anything fitted on a distribution is deferred to the pre-modelling stage (Section 6). The columns added here are: the end-of-surgery timestamp derived from the transfer time out of the surgical bed, the binary surgical-drain indicator, and the measurement-taken indicators created for the numeric and binary fields in which absence is clinically legitimate. The cleaned table handed to feature engineering therefore contains 31,849 surgical events and 97 columns.

Finally, the entire exploratory analysis was re-executed on the cleaned dataset and the two runs were compared side by side, so that any change in missingness, in duplicate count or in outcome prevalence introduced by cleaning would be visible rather than silent.

Enforcement of the prediction window ran as a distinct step before any other cleaning decision, so that the question of whether a value may exist at prediction time was settled before the question of how it should be represented; the two mechanisms it uses - the unconditional drop list and the empirical timing gate - are described earlier in this section.

This separation matters because the two mechanisms answer different questions. The keyword audit identifies candidates for review, whereas the timing gate is an empirical test applied to the actual timestamps of each record. Neither is a feature-engineering decision: at this stage the pipeline determines only which values may legitimately be seen at the moment of prediction, and every subsequent representational choice is made in the Feature engineering section on the surviving columns. The guiding principle behind the pipeline as a whole is the strict separation of deterministic, clinically defined cleaning transformations from data-dependent statistical learning, so that the held-out test set is protected and data leakage is prevented by construction rather than detected after the fact.

4. Clinical text processing

The dataset is not purely tabular. Narrative fields contain clinical information that is either absent from the structured columns or represented there only as a coded conclusion. Discarding them would discard exactly the part of the record in which a developing clinical picture is described in words. Three narrative fields were examined and two were carried into the text

pipeline: the recorded medical history of the admission and the description of the operative course. The third, the main complaint on admission, was excluded after the textual exploratory analysis showed that its content is largely reproduced inside the medical-history field, with a mean similarity ratio of 0.30 and 15% of sampled rows overlapping heavily; passing both would have supplied the same words twice. The post-operative assessment field was excluded for a different and stronger reason: it is recorded after the prediction point and is therefore inadmissible under the window fixed in the Definition of the prediction target and prediction window section. The past-surgery description and the preoperative regular-medication list were also kept out of the text pipeline: their content is short and list-like rather than narrative, so a free-text risk score would add little beyond what can be extracted deterministically, and they are represented instead by presence flags and clinically approved medication-family flags during feature engineering (Section 5).

The pipeline begins with a textual exploratory stage, because the choice of language model depends on what language the notes are actually written in rather than on what they are assumed to be written in. The character composition of every note was measured. In the medical-history field 75.8% of notes are Hebrew only and 23.7% mix Hebrew with Latin characters, with a median length of 26 words and a maximum of 1,554; in the operative description 60.4% are Hebrew only and 20.5% mixed, with a median of 66 words. Both fields contain substantial repetition - 12.6% and 28.8% of values respectively are exact duplicates of another note - which is characteristic of template-based operative documentation and is one reason a term-frequency representation performs as well as it does here. Missingness is itself informative: the medical-history field is absent for 10.4% of non-infected admissions but for 33.3% of infected ones, while the operative description is present for every case.

A layer of interpretable, rule-based features is extracted next. The notes are segmented into their labelled Hebrew sections, ages and dates mentioned inline are extracted by regular expression, the text is lightly normalised by collapsing whitespace and stripping stray punctuation while preserving Hebrew characters, and a curated comorbidity dictionary is applied. Coverage was measured before use rather than assumed: an inline age is recoverable for 33.1% of notes, and the most frequently detected comorbidity families are malignancy (10.6%), cardiac disease (5.1%), hypertension (4.8%), obesity (4.4%), smoking (2.8%) and respiratory disease (2.7%), with diabetes, renal failure, immunosuppression and transplantation each below 2%. These features are transparent and cheap, and they serve as a sanity check on the learned representations rather than as the primary text signal.

Two frequency-based baselines were then implemented on a single shared evaluation split so that every text condition is judged on the same cases. Each of the two narrative fields is vectorised by its own TF-IDF vectoriser inside a column transformer, over unigrams and bigrams, limited to 1,500 features per field, requiring each term to appear in at least two documents, and with a 194-word Hebrew stop-word list loaded from a local package. Keeping the vectorisers separate prevents the longer operative description from dominating the

representation and preserves the field-specific vocabulary of each note type. The first baseline classifies the resulting matrix with class-weighted logistic regression, the second with an XGBoost classifier whose positive-class weight is derived from the training data, so that the comparison isolates the effect of the classifier rather than of the representation.

The advanced condition was originally planned as end-to-end fine-tuning of AlephBERT, a Transformer pre-trained on a large general Hebrew corpus. In practice this proved impractical inside the hospital computing environment: no suitable GPU was available, and fine-tuning a model of that size on a few hundred positive cases carried a substantial overfitting risk. The advanced condition was therefore re-specified as frozen inference against a large language model served locally in the hospital's Ollama environment. Several locally available models were tested: llama4:maverick did not return responses within a practical inference time and repeatedly timed out, and after consultation with the hospital technical team qwen2.5-coder:32b and llama3.3:70b were compared on small smoke tests. Both returned well-formed structured output, and llama3.3:70b was selected because its general natural-language capabilities suit the interpretation of Hebrew clinical narrative better than a model optimised primarily for programming tasks. Each surgical case is presented to the model as a structured prompt in which the two eligible fields - `first_hospitalization_disease_history` and `description_surgical_procedure` - appear as separately labelled sections, at temperature zero, and the model is asked for a JSON object. The prompt is written deliberately to ask for forward-looking risk factors and early warning signs rather than for confirmation of an infection already documented in the text, because a model that merely recognises a recorded diagnosis is "reading the answer off the page" rather than predicting anything.

The model is asked for a continuous risk score in addition to interpretable flags, rather than for a high or low label alone; the Low/High classification it also returns is retained only as an auxiliary representation and is not used as a predictor. A binary label discards information: two records classified as high risk may carry scores of 0.52 and 0.94, and for a downstream tree-based model that difference may be informative. Inference was executed over the whole cleaned cohort with persistent caching and periodic checkpoints keyed by the surgical identifier, so that an interrupted run resumes without recomputing completed cases. A row whose call failed, or which returned valid JSON without a usable score, is cached with a missing score and excluded from evaluation rather than being silently defaulted to the negative class; across all 31,849 cases no such failure occurred.

The three conditions were compared on the shared evaluation split, and the frequency-based baseline won decisively. That comparison is the output of this stage. The consequence for the architecture is that the representation carried into the tabular model is the logistic-regression score over the TF-IDF matrix - refitted and generated out of fold in the following stage, as described in the Feature engineering section - and not the language-model score. The cached language-model outputs, the continuous score together with six structured clinical indicator flags, are retained as a reference table only, so that the alternative feature set remains available

for the future work proposed in the Recommendations for further research and development section without taking any part in the results reported here.

Transferring a text-derived score into a tabular model creates a leakage risk of its own, because a score produced by a model that was trained on the same case already contains information about that case's outcome. The generation procedure is therefore constrained in two ways, and this is described in full in the Feature engineering section because it is executed at the start of the feature-engineering stage. The train/test split is created and saved to disk first, so that every later notebook loads the identical partition; training cases are then scored out of fold, each case by a model fitted on the other folds only, with the TF-IDF vectorisers refitted inside each fold so that no vocabulary or inverse-document-frequency information crosses the fold boundary; and the held-out test cases are scored once by a model fitted on the complete training partition. The resulting column is merged into the structured feature table on case_number, after validating that the text table contains exactly one row per surgical case, and every case is asserted to carry a score before the table is saved.

Two properties of this arrangement should be stated precisely, because they bound what the text-derived feature is. First, the score that enters the tabular model is not produced by the exact pipeline that won the comparison of the Clinical text models section. That comparison used a deliberately constrained representation - one vectoriser per field, unigrams and bigrams, at most 1,500 features per field, a minimum document frequency of two and a 194-word Hebrew stop-word list - chosen to control dimensionality on a small positive class. The score generated in the feature-engineering stage uses the same architecture, two separate vectorisers per field feeding one class-weighted logistic regression, but with the vectorisers at their library defaults: unigrams only, no feature cap, no minimum document frequency and no stop-word list. The consequence is that the feature carried forward is a summary produced by a related but not identical text model, and the average precision of 0.519 reported in the Clinical text models section is therefore not an estimate of how much that particular column can contribute.

Second, the split on which the three text conditions were compared is not the modelling split. It is a separate, deliberately rebalanced partition drawn from the full export before the cohort restriction - 35,606 notes for training and 319 for evaluation, constructed to hold one fifth of the positive notes against ten negative notes for each positive one - and it therefore overlaps both modelling partitions. That construction is appropriate for its purpose, which is to rank three representations against one another on the same cases, and it is the reason the figures in the Clinical text models section are described there as relative merit rather than as performance on the surgical population. It does, however, mean that the choice of representation was informed by notes belonging to cases that later appear in the frozen modelling test partition. No parameter crosses that boundary - the score itself is generated out of fold against the locked split - but the architectural decision was not made blind to it, and this is recorded as a limitation in the Limitations section.

Keeping the text output as a feature rather than as a final classifier is what makes the central comparison of this project possible: the same locked algorithm can be trained once on the structured features alone and once on the structured features together with the text-derived score, so that the contribution of the narrative is measured as an increment over the tabular model rather than as a separate competing model. That comparison is reported in the Multimodal contribution, explainability and error analysis section.

5. Feature engineering

Feature engineering is distinct from cleaning: it does not correct errors but represents existing clinical information in a form a model can learn from. Every deterministic transformation in this stage is defined by a clinical or logical rule and none of them estimates a parameter from the data distribution, which is why they may be applied to the complete dataset. The one exception is the text-derived risk score, which is a learned quantity: it is therefore generated only after the train/test split has been created and saved at the very start of this stage, out of fold on the training cases and by a fully fitted model on the test cases, exactly as described in the Clinical text processing section. The objective of the stage is a semantically complete feature table, after which all remaining preprocessing is learned from the training partition alone.

An indicator of direct transfer to intensive care (`transferred_to_icu_directly`) is derived from the raw ICU-admission timestamp relative to the calculated end of surgery, using a six-hour tolerance window: it is set to one only when the ICU record falls on or after the calculated end of surgery and within that window, and to zero otherwise. Zero therefore covers three distinct situations, which the audit reports separately even though the stored value is the same: an ICU transfer recorded before the end of surgery, which is implausible as a direct post-operative transfer (247 events); an ICU transfer recorded too long afterwards; and no ICU record at all (29,191 events), which in this hospital means the patient did not enter intensive care, because clinicians document the transfer when it occurs. The feature is positive for 2,411 surgical events. This is a clinically defined encoding of absence, and the raw timestamp and bed identifier are dropped once it has been computed.

Death during admission was not used as a predictor. Only one surgical event in the cohort involved death during the same admission. Because a patient who dies early may not have had enough time to develop or document the infection outcome, this case was reviewed separately as a potential competing-risk case. Following clinical review, it was retained in the cohort and was not excluded from the analysis.

The remaining deterministic features were selected from patterns observed during the exploratory analysis of infected patients, while deliberately avoiding rules defined against the outcome itself. Baseline physiological burden is represented by a categorical body mass index on the World Health Organization bands, alongside the retained continuous value, and by a binary indicator of ASA class 3 or above, which is positive for 5,665 events. A binary mental-

health indicator is derived from the free-text preoperative mental-health field. The payer field, whose raw values are a long tail of insurer names, is collapsed into nine interpretable categories distinguishing the public sick funds, private insurance, foreign insurance, medical tourism, public-hospital funding and self-payment. Perioperative invasive devices are represented by three flags derived from the timing-gated device records - a central venous catheter (2,187 events), a urinary catheter (4,294) and a surgical drain left in place (1,689) - together with a count of the devices present. Two duration features are computed from the transfer timestamps: time in the operating-room bed and time in the recovery bed, with negative durations masked as missing rather than clipped to zero.

One feature was deliberately not created, and the reasoning is recorded here because its absence is a decision rather than an omission: an age threshold indicator was not generated, because the cohort restriction already removes the younger population (see the Cohort definition and unit of analysis section) and the continuous age variable is retained in full. Two compact documentation-burden counts were derived from the diagnosis and procedure fields: the number of documented diagnoses and the number of documented procedures.

The field `preop_regular_medications` lists several medications within a single cell. One-hot encoding the cell as a whole would create spurious categories corresponding to particular drug combinations rather than recognising the individual components. The field is therefore parsed into a medication vocabulary with frequencies, trade names and spelling variants are consolidated into drug families, and binary flags are produced only for families that were clinically approved. The raw text is removed only after the derived features have been generated, so that no information is lost before its replacement is verified.

Four medication-family features were approved and generated from the parsed vocabulary: statin use (5,907 events), antiplatelet use (3,421), glucose-lowering therapy (2,747) and chronic systemic corticosteroid use (130).

These medication families were selected on the basis of clinical relevance rather than frequency alone. Glucose-lowering therapy was included because it serves as a marker of diabetes or impaired glucose regulation, both of which are associated with perioperative hyperglycaemia and impaired wound healing. Chronic systemic corticosteroid use was retained because prolonged steroid exposure may suppress immune function and delay tissue repair. Antiplatelet therapy was included because of its potential association with perioperative bleeding and haematoma formation, which may adversely affect wound healing. Statin use was retained because it is common among patients with cardiovascular and metabolic comorbidities and may also exert immunomodulatory and anti-inflammatory effects that could be relevant to postoperative infection risk.

Each feature is set to one when at least one validated medication belonging to the corresponding family is identified in the preoperative medication field, and to zero otherwise. Only explicitly reviewed medication names are included; ambiguous tokens, dosage terms, administration

instructions and uncertain abbreviations were excluded rather than classified automatically. The corticosteroid feature was retained despite its low prevalence because chronic systemic steroid exposure represents a biologically plausible and clinically important risk-related exposure rather than a frequency-driven predictor. In a retrospective analysis of 403,566 arthroplasty patients in the ACS-NSQIP registry, patients on chronic corticosteroid regimens had significantly higher rates of superficial, deep incisional and organ-space surgical site infection as well as wound dehiscence [20].

Surgeon and anaesthetist identifiers were audited rather than removed in advance, and the audit decided against them. The surgeon field holds 371 distinct codes, of which 129 appear in fewer than five surgical cases; the anaesthetist field holds 190, of which 90 appear in fewer than five. A high-cardinality identifier of this kind generates a large number of sparse categories with too little statistical support to learn a stable association, and it encodes the identity of an individual clinician rather than an intrinsic characteristic of the patient or the procedure, so a model could learn local referral and case-mix patterns instead of generalisable clinical risk. Both identifiers were therefore excluded from the model features, while the more generalisable surgical specialisation and activity cluster were retained. The feature table handed to the pre-modelling stage contains 31,849 surgical events and 92 columns.

No cohort reduction, class rebalancing or undersampling is performed at this stage. Those are sampling decisions rather than representational ones, and applying them before the data are partitioned would allow the composition of the test set to be influenced by choices made for the benefit of training; they are therefore deferred to the pre-modelling stage (Section 6). The stage concludes by saving a single final feature table, in which the raw narrative text and the confirmed redundant source fields have been removed only after their derived features were created and verified, so that the partitioning stage receives one semantically complete and fully documented input.

6. Pre-modelling, class imbalance and data partitioning

From this stage onwards every operation is capable of being influenced by the distribution of the data, and therefore nothing is fitted before the population entering the models has been fixed and the data have been partitioned. The stage opens with a concise final check of the outcome variable - the count of positive and negative cases, the prevalence, the imbalance ratio and the absence of missing or non-binary values.

The single most consequential property of that cohort for the modelling design is the severity of its class imbalance. The restricted cohort comprises 31,849 surgical events, of which 141 are positive for infection - a prevalence of 0.443%, or approximately 225 negative cases for each positive one. A conventional proportional split holding out 30% of the data therefore places only about forty positive cases in the test partition, which is a very thin basis for any estimate, and it presents the training algorithms with a class the optimiser can essentially ignore while still achieving above 99% accuracy.

Two distinct kinds of decision address this, and they are kept strictly separate because they answer different questions. The first is the Final Study Cohort restriction described in the Cohort definition and unit of analysis section: a population-definition decision about which surgical events this model is declared to address at all, applied once to the complete dataset during cleaning and therefore before the split, so that both partitions describe the same population. The second is a rebalancing decision about what the training algorithm is shown, which may never touch the test partition.

The train/test split itself is not created in this stage. It was created once, at the start of the feature-engineering stage, as a stratified 70/30 partition of the restricted cohort with a fixed random seed, and written to disk as an explicit list of case identifiers with their assigned partition; it is loaded unchanged here and validated to cover every case exactly once. A 30% test fraction was preferred to the more usual 20% because of the class imbalance: with 141 positive events in the cohort, an 80/20 split would leave roughly 28 positive cases in the test partition, whereas 70/30 leaves 42. Every additional positive case materially stabilises recall and precision estimates computed over so few events, and with more than twenty-two thousand training rows the ten percentage points given up are not the binding constraint on what the model can learn. The training partition holds 22,294 surgical events with 99 positive cases (0.444%) and the frozen test partition holds 9,555 events with 42 positive cases (0.440%), so the positive rate is preserved in both. Creating the split before the text-derived score was generated is what allows that score to be produced out of fold on the training cases and by a fully fitted model on the test cases, without the partition being redefined afterwards.

Only once the partition has been loaded is any rebalancing allowed to touch the data, and only the training partition is ever affected. The technique adopted is prototype, or aggregation, undersampling of the negative class, adopted as an alternative to deleting majority-class rows at random: groups of similar negative patients are replaced by a single representative row, so that the information they carry is summarised rather than discarded. Similarity is defined by clinically meaningful strata - age band, the high-ASA indicator, body-mass-index category, activity cluster and sex - and within each stratum every 150 negative rows become one representative row, independently of how many positive cases that stratum contains, including strata with none. Strata holding fewer than 150 negative rows, and any remainder that does not fill a complete group, are kept as individual rows. Aggregation is type-aware rather than a blind average: genuinely continuous measurements take the median, integer-valued measurements take the median rounded to the nearest whole number so that a representative row never carries an impossible value such as two and a half diagnoses, and binary and categorical fields take the mode. Positive cases are never grouped, never averaged and never removed. The procedure reduced the training partition from 22,294 rows to 8,288 - 8,189 negative and all 99 positive - raising the prevalence the model is shown from 0.444% to 1.19%, while the frozen test partition retains the true cohort prevalence. The test partition is deliberately left untouched by any sampling decision, because it is the only object in the pipeline that still represents the clinical

reality the system would face in use: an infection-prevention team screens a surgical population in which roughly four patients per thousand develop an infection, and precision, the alert rate per 1,000 procedures and the number needed to screen are meaningful only when measured at that prevalence. Rebalancing the test set would raise the apparent precision of the model without changing anything on the ward, and would therefore overstate the clinical value of the system to the people who must act on its alerts. For the same reason the partition is written to disk as a fixed list of case identifiers and is neither regenerated nor re-stratified at any later stage. Full balance was rejected deliberately: a training set in which half the patients are infected describes a population that does not exist.

All further data-dependent preprocessing is then fitted on the training partition only and applied unchanged to the test partition. Categorical values occurring in fewer than 3% of training rows are grouped into a single residual category, and the learned list of rare values is applied to the test set as it stands, so that no test frequency influences the grouping. The 43 categorical predictors are then one-hot encoded by an encoder fitted on the training partition, producing 214 indicator columns and ignoring any category seen only in the test set; a technical missing category is used for categorical fields that legitimately remain empty after cleaning. Numeric predictors missing for more than 90% of training rows are dropped, since a median cannot rescue them; those missing for more than half receive an explicit missingness indicator alongside the imputed value, so that the fact of absence remains available to the model; and the remaining gaps are filled with the training median. The resulting matrices are fully numeric and complete, with 266 columns, 8,288 training rows and 9,555 test rows.

Two further reviews close the stage, and both produce evidence rather than automatic deletions. The variance inflation factor is computed over the original numeric predictors and identifies exact linear dependence among the invasive-device features, where the device count is by construction the sum of the three device flags, and very high values for the intra-operative temperature and sodium measurements; no feature is removed automatically, because in a clinical model redundancy is sometimes the price of interpretability, but the finding is what justifies preferring a regularised linear model to an unregularised one. Feature-selection evidence is then produced in two complementary ways: mutual information, which can capture non-linear association, and the coefficients of an L1-regularised logistic regression fitted on standardised features, which shrinks weak contributions to exactly zero. The two rankings are combined into a mean rank, and the twenty highest-ranked features were carried into the modelling stage. This is a deliberate reduction: with 99 positive training cases, a 266-column matrix invites overfitting, and a compact feature set is also what makes the coefficient and SHAP explanations of the Multimodal contribution, explainability and error analysis section legible. Its cost is discussed as a limitation in the Limitations section.

7. Modelling strategy and hyperparameter tuning

The modelling stage was not organised as a single competition in which several algorithms are run and the highest number is selected. Because of the extreme class imbalance and the

computational cost of the balancing techniques considered, the decision was decomposed into four separate questions: how the training set should be balanced, which algorithm performs best under that balancing strategy, at which operating threshold it should be applied, and whether the probabilities it produces are trustworthy.

A DummyClassifier baseline is evaluated first. Where fewer than one per cent of patients develop an infection, a model that always predicts the negative class achieves very high accuracy while carrying no clinical value whatsoever; the dummy baseline makes that arithmetic explicit and provides the reference against which the real models must demonstrate genuine predictive ability rather than a reproduction of the base rate.

Evaluating every algorithm under every balancing strategy would have required many full tuning runs, which was not computationally feasible, so balancing selection was separated from hyperparameter tuning. The training partition entering this stage has already been through prototype aggregation, so the screen does not ask whether rebalancing is necessary in the abstract; it asks whether anything should be added on top of the prototypes. The baseline condition is therefore the prototype-aggregated training set used as it stands, against which three additions were compared: automatic class weighting, SMOTE oversampling of the minority class, and SMOTENC, the variant of SMOTE that treats categorical columns explicitly. Each strategy is evaluated with two representative models, logistic regression and random forest, under three-fold cross-validation with no hyperparameter search, and the resampling is performed inside the fitting step so that only the training portion of each fold is ever augmented. Two synthetic-data strategies specified in the plan, CTGAN and CTGAN combined with SMOTE, were excluded from automatic selection by design: the matrices reaching this stage are already one-hot encoded, and a generator that treats each indicator column independently can produce combinations that correspond to no valid original category, such as a record belonging to two mutually exclusive ASA classes at once. They could not have been run in this environment in any case, since the package is not installed and no external installation is possible.

The two decisions were therefore taken in sequence rather than jointly. The balancing strategy with the best mean rank across the two representative models was locked first, and only then were the four candidate algorithms - logistic regression, random forest, XGBoost and CatBoost - tuned in full.

The reason for this ordering is arithmetic. A joint search would have to explore the hyperparameter space of every algorithm inside every balancing condition, so its cost is the product of the two rather than their sum: four balancing conditions by four algorithms by twenty sampled configurations by five cross-validation folds, with the resampling regenerated inside every fold, is well beyond what the available hardware could complete in the time available. Separating the decisions reduces the work to one inexpensive screen - two fixed models, no hyperparameter search, three folds - followed by a single full tuning run per algorithm under the locked strategy.

The assumption this makes is that the ranking of balancing strategies is broadly stable across algorithms. It is stated here because it is an assumption rather than a result, and it is the reason the screen deliberately uses two models of different character, one linear and one tree-based, and selects on mean rank rather than on the single best score. The residual risk is that an algorithm not represented in the screen might have performed better under a different balancing strategy; this is accepted as the cost of feasibility and is recorded among the limitations in the Limitations section.

With the strategy fixed, each model is tuned by randomised search over twenty candidate configurations with five-fold stratified cross-validation, optimising average precision, and the search is nested inside the resampling wrapper so that synthetic minority cases are generated separately for each fold and each configuration. The final algorithm is selected by the highest cross-validated average precision on the training partition, before the test set is touched at all.

Each model is tuned with RandomizedSearchCV. Hyperparameters are the settings that govern how a model learns - tree depth, learning rate or regularisation strength, for example - and they are not selected by inspecting test-set performance. Candidate combinations are searched on the training set under cross-validation only and ranked by PR-AUC, so that the test set plays no part in model selection.

PR-AUC was chosen as the ranking criterion rather than accuracy, ROC-AUC or F1 because it focuses directly on performance for the rare positive class and on the trade-off between precision and recall. With roughly four positive cases per thousand, accuracy can remain extremely high even for a model that predicts almost every case as negative. ROC-AUC remains useful as a measure of discrimination, but in a highly imbalanced population it can appear favourable even when the model produces a clinically unacceptable number of false-positive alerts. F1, in contrast, depends on a specific decision threshold selected in advance. PR-AUC summarises precision across the full range of recall values without fixing that operating threshold, making it more appropriate for comparing candidate models during tuning. The final clinical threshold is therefore selected separately in the Evaluation, threshold tuning, calibration and explainability section.

8. Evaluation, threshold tuning, calibration and explainability

Because infection cases are rare, accuracy is a deceptive measure and was not allowed to drive any decision. PR-AUC was adopted as the primary selection criterion because it reflects directly the trade-off between identifying true infections and limiting false alerts. ROC-AUC is also reported, but with the caveat that a very large number of true negatives can make it appear favourable even when the false-alert burden is high. Recall, precision, F1, F2, balanced accuracy and the number of alerts generated per 1,000 procedures are reported alongside it, so that the evaluation describes both the statistical quality of the model and its operational cost to an infection-prevention team.

The decision threshold was tuned rather than assumed against the number needed to screen (Formula 2). A model returns a probability, and converting that probability into a binary alert requires a cut-off; 0.5 is a mathematical default, not a clinical decision. In this application a false negative - a patient who develops an infection that the model failed to flag - is more costly than a false positive, so a wide range of thresholds is evaluated on out-of-fold training predictions, reporting recall, precision, F1, F2, false-positive and false-negative counts, the number needed to screen and the alert rate per 1,000 procedures at each. The number needed to screen is simply the inverse of precision,

$$\text{NNS} = 1 / \text{precision} = (\text{TP} + \text{FP}) / \text{TP}$$

Formula 2. Number needed to screen. TP and FP are the true- and false-positive counts at a given threshold; NNS is the number of flagged surgical events an IPC nurse must review in order to find one true infection, and is the quantity against which the criterion agreed with the hospital was stated.

The selection rule follows a criterion set following clinical consultation with the hospital: a workload of screening approximately twenty flagged cases in order to find one true infection is acceptable. The threshold is therefore chosen to maximise recall among all thresholds whose number needed to screen does not exceed twenty, which is equivalent to requiring a precision of at least 5%. Applied to the out-of-fold training predictions of the locked model, this selects a threshold of 0.200, at which the model reaches 92.9% recall with a precision of 5.2% and a number needed to screen of 19.3. The same rule is reapplied without modification after calibration, so that calibration cannot silently replace the approved clinical criterion; an F2-optimal fallback exists in the code but was not used, because the clinical criterion was available.

Probability calibration is evaluated separately, because good discrimination does not guarantee trustworthy probabilities: a model may rank patients correctly and still assign a risk of 0.80 to a group in which only a small minority develop an infection. Three options - no calibration, sigmoid (Platt) scaling and isotonic regression - were compared on the training partition alone by Brier score, with the calibrator fitted inside cross-validation so that no case contributes to both its own calibrator and its own calibrated prediction. Isotonic regression was selected, at an out-of-fold Brier score of 0.0098 against 0.0113 for sigmoid scaling and 0.1032 for the uncalibrated model. Because isotonic regression changes the probability scale substantially - the uncalibrated logistic model is pushed towards the center of the range by the rebalanced training set, whereas the calibrated model returns probabilities near the true prevalence - the threshold was re-selected on the calibrated out-of-fold predictions under the same clinical rule, giving a final locked threshold of 0.010 at 90.9% out-of-fold recall, 5.6% precision and a number needed to screen of 17.9. A low threshold is not a sign of a weak decision rule here; it is the arithmetic consequence of a calibrated probability scale on a population in which fewer than one patient in two hundred develops an infection.

The test set is used only after the balancing strategy, the algorithm, the hyperparameters, the calibration method and the threshold have all been fixed. Selecting any of these on the basis of test-set performance would mean that the test set is no longer an independent estimate of

generalisation. The final evaluation reports PR-AUC, ROC-AUC, recall, precision, F1, F2, balanced accuracy, the Brier score, the confusion-matrix counts and the alert burden per 1,000 procedures, and is accompanied for each model by a confusion matrix, a ROC curve and a precision-recall curve.

Explainable AI is treated as a requirement of the system rather than as a presentational addition, because in a clinical setting a risk score of 0.78 is not actionable unless the variables that produced it can be identified. Three layers of explanation are therefore produced: coefficients and odds ratios for the logistic regression, permutation importance across models, and SHAP values for the final selected model. SHAP provides a global view of which features influence the model most and a local view of which features pushed an individual patient's prediction towards higher or lower risk; to keep the computation tractable it is evaluated on a bounded sample of the test set rather than on every record. The literature review explicitly identified SHAP as a means of showing clinicians which variables drove a decision [12], and the project proposal identified interpretability as a risk to be mitigated through explainable AI.

Finally, the evaluation does not stop at aggregate metrics. False negatives and false positives are retained by case_number, because in a clinical model an average metric does not tell the whole story: failure cases can reveal missing patterns, specific procedure types, patient groups that are intrinsically difficult to predict, or residual leakage that was not identified earlier. Subgroup performance is additionally examined where the group sizes permit - for example by ASA classification, sex and department - not in order to re-tune the model on the test set, but to establish whether it behaves consistently across the population it would serve.

Problems encountered and their resolution

Several methodological, technical, and clinical challenges arose during the execution of the project and required adaptations to the original workflow. The main difficulties and the solutions implemented are described below.

1. Delayed access to the final dataset: Access to the complete hospital dataset was granted later than originally expected. As a result, several initial methodological decisions had to be made before the data could be examined directly, including the choice of candidate model families, evaluation metrics, and strategies for addressing class imbalance. During this period, the work focused on the literature review, clinical characterization, and preparation of the analytical framework. Once access to the dataset was granted, these initial decisions were reassessed and adapted according to the actual data structure and observed distributions.

2. Construction of the infection outcome: Infection status was not provided as a single complete binary variable and therefore had to be derived from two diagnosis-related fields that were not populated consistently across all records. A unified labelling rule was therefore defined, according to which a case was classified as positive when at least one of the two relevant fields

indicated a documented postoperative infection. This rule was applied consistently throughout the entire analytical pipeline.

3. Extreme class imbalance: One of the most substantial challenges in the project was the very low prevalence of postoperative infection, with only 141 positive cases among 35,925 surgical records. This limited the number of positive examples available for both model training and evaluation and made conventional measures such as accuracy unsuitable as primary performance metrics. The response included stratified data partitioning, application of balancing strategies only within the training data, use of evaluation metrics appropriate for rare outcomes such as PR-AUC and recall, and threshold selection that considered the expected clinical alert burden rather than relying on the default threshold of 0.5.

4. Refinement of the prediction time point: The initial project plan considered the possible use of information from the early postoperative period. However, exploratory analysis showed that some postoperative variables were documented only after clinical suspicion of infection had already emerged. Including such variables would have blurred the distinction between early prediction and retrospective detection. The prediction point was therefore refined to the end of the surgical procedure, and the eligible feature set was redefined according to whether each variable was genuinely available at that time.

5. Prevention of temporal data leakage: The retrospective structure of the dataset included variables recorded at different stages of the postoperative course, some of which would not have been available at the intended prediction time. Their inclusion could have artificially inflated model performance. A structured leakage-prevention process was therefore applied: variables known to occur after the prediction window were excluded, while time-dependent variables were retained only when their timestamps confirmed that the information had been available within the permitted time window. This process reduced the usable feature set but was necessary to preserve the methodological validity of the prediction task.

6. Constraints on advanced Hebrew clinical NLP: The clinical narratives were written predominantly in Hebrew, while all analysis had to be performed within the hospital's secure computing environment. This combination limited access to external NLP resources and to the computational infrastructure required for training large language models. The original plan included fine-tuning an advanced Hebrew language model, but this was not feasible with the available infrastructure. The NLP strategy was therefore adapted to combine classical TF-IDF-based models with locally hosted LLM inference through the hospital's Ollama environment, without fine-tuning. In addition, some locally available models were not practical because of long inference times and repeated request timeouts. Small smoke tests were therefore performed on alternative models, and a locally available general-purpose model was selected as the more practical option for interpretation of Hebrew clinical narratives.

Expected outcomes

Based on the methodology described above, the project was expected to produce a clinically interpretable framework for the early prediction of postoperative surgical site infection using routinely collected electronic health record data. The expected outcomes were defined as project deliverables and analytical findings rather than as predetermined levels of model performance, since the achievable predictive performance depended strongly on the quality of the available data and on the extremely small number of positive infection cases.

1. An evaluated predictive model for early postoperative infection risk:

The primary expected outcome was a machine-learning model capable of estimating the risk of postoperative infection at the defined end-of-surgery prediction point. The model was expected to provide meaningful discrimination between higher- and lower-risk surgical cases while being evaluated with metrics appropriate for the highly imbalanced outcome, particularly precision, recall and PR-AUC.

2. Identification of clinically relevant predictors of infection risk:

The project was expected to identify which demographic, preoperative, intraoperative and procedural characteristics contribute most strongly to infection-risk prediction. This analysis was intended both to improve the predictive framework and to provide clinically meaningful insight into the factors associated with postoperative infection in the study population.

3. Assessment of the added value of clinical free text:

Because important clinical information may be documented only in narrative notes, the project was expected to determine whether information extracted from Hebrew clinical text provides additional predictive value beyond structured tabular variables alone. The multimodal comparison was therefore intended to quantify whether incorporating an NLP-derived representation improves prediction sufficiently to justify its inclusion in the final framework.

4. A clinically usable screening strategy:

Rather than producing a binary prediction based solely on the conventional probability threshold of 0.5, the project was expected to establish an operating point that balances the detection of infected patients against the number of false-positive alerts generated. The intended outcome was therefore a screening framework that could prioritise a manageable subset of surgical cases for closer clinical review by infection-prevention personnel.

5. An interpretable and transparent prediction framework:

The final modelling framework was expected to include explainability methods that identify the variables contributing to both overall model behaviour and individual predictions. This was intended to support clinical interpretation, facilitate detection of implausible model behaviour or residual data leakage, and improve the transparency required for potential future clinical use.

6. A reproducible analytical pipeline suitable for future validation and development:



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In addition to model performance, the project was expected to deliver a documented and reproducible workflow covering cohort definition, target construction, prediction-window enforcement, data cleaning, feature engineering, NLP processing, class-imbalance handling, model training, calibration, threshold selection and explainability. Such a pipeline would provide a basis for future validation on additional data and for subsequent development toward a clinical decision-support tool.

The extent to which these expected outcomes were achieved is evaluated in Chapter 5 and discussed in relation to the original project goals in Chapter 6.

Project Results

Cohort and exploratory findings

The exploratory analysis described in the cohort-definition and exploratory stages above confirmed a raw cohort of 35,925 surgical hospitalisations, of which 141 (0.392%) carried a positive infection label. The extreme rarity of the positive class, rather than any single variable, was the finding that most strongly shaped the subsequent modelling strategy. Table 2 traces the cohort from the raw hospital export to the training and test populations used for modelling.

Table 2. Cohort reduction from the raw export to the modelling tables. Prototype aggregation increased the prevalence presented to the model in the training partition from 0.444% to 1.19% without removing any positive case. In contrast, the frozen test partition was left completely unchanged and retained the true restricted-cohort prevalence of 0.440%.

Stage	Events	Positive	Prevalence	Why
Raw export as received	35,925	141	0.392%	All surgical events supplied by the medical center
After Final Study Cohort restriction	31,849	141	0.443%	Age under 28 and four frozen zero-infection activity clusters removed (Cohort definition and unit of analysis)
Training partition (before rebalancing)	22,294	99	0.444%	Stratified 70/30 split of the restricted cohort, fixed seed, created during feature engineering
Training partition after prototype aggregation	8,288	99	1.19%	Groups of 150 similar negative cases replaced by one representative row; no positive case touched
Frozen test partition	9,555	42	0.440%	Never rebalanced, never used for any selection decision

This distinction was essential: rebalancing supported model learning only, whereas the test partition had to preserve the clinical population on which generalisation would ultimately be assessed. Because the split was stratified, the positive prevalence remained nearly identical across the original training and test partitions.

The rank-biserial correlation is reported as a non-parametric effect-size measure, with larger absolute values indicating stronger separation between infected and non-infected admissions. The strongest univariate associations with the infection outcome were initially observed in variables that were themselves consequences of infection rather than true predictors of future risk. Total length of stay had a median of 185 hours among infected admissions compared with 23 hours among controls (rank-biserial $r = 0.87$). Readmission length of stay, the number of blood or wound cultures obtained between two and ninety days after surgery, and drain and urinary-catheter durations followed the same pattern ($r = 0.73, 0.54, 0.49$ and 0.33 respectively).

These variables were recorded after the defined prediction point and were therefore excluded through the leakage-control procedures described in the Data cleaning and prevention of data leakage section. They are reported here only because their strength is itself the methodological finding: a model permitted to use them would appear highly discriminative while in fact exploiting information that becomes available only after clinical suspicion or infection development has already occurred.

Among variables genuinely available at or before the end-of-surgery prediction point, the strongest clinically plausible associations are summarised in Table 3.

Table 3. Strongest associations among variables available at the prediction point. Only variables whose values were established at or before the prediction point are included in this table. Measurements that mature only later in the postoperative course are excluded from the table and from the feature set, however strong their association with infection.

Variable	Infected (median)	Not infected (median)	Rank-biserial r
Time in the operating-room bed (hours)	3.68	1.28	0.81
Recorded operative duration (minutes)	164	80	0.63
Minimum intra-operative temperature ($^{\circ}\text{C}$)	35.1	36.1	-0.42
Age (years)	65.1	55.5	0.34
Body mass index	28.05	26.54	0.21

The largest eligible univariate association was observed for time spent in the operating-room bed, followed by recorded operative duration, minimum intra-operative temperature, age and body mass index. All of these associations remained statistically significant after Benjamini-Hochberg correction across the full set of bivariate tests. The effect directions were also clinically plausible: infected patients tended to undergo longer procedures, had longer

operating-room exposure, were older and had higher body mass index, while lower minimum intra-operative temperature was associated with infection.

A dedicated temporal analysis further clarified which measurements could legitimately be used at the selected prediction point. Laboratory testing was predominantly postoperative: only 7.4% of first laboratory results were available before surgery, while the median first panel was recorded approximately 15 hours afterwards and approximately 81% fell within the first 24 postoperative hours. Microbiological events occurred considerably later. Blood cultures were recorded at a median of approximately 93 hours after surgery, and wound or additional cultures at approximately 187 hours. They were also present in only a small fraction of surgical records. This pattern supports the conclusion that their presence often reflects an already established clinical reason for investigation rather than information available for prospective prediction.

In contrast, several perioperative variables occurred sufficiently close to surgery to remain eligible. Central-line insertion, for example, was recorded at a median of approximately 0.35 hours after the end of surgery, with the large majority occurring within the allowed time window. This explains why the central-line group passed the timing gate described in the Data cleaning and prevention of data leakage section, whereas later interventions such as parenteral nutrition did not.

Multivariate analysis of the raw tabular dataset, as received from the hospital and before any cleaning, also identified substantial redundancy between several variables, including readmission status and readmission date, repeated laboratory measurements, and multiple descriptions of the same bed-transfer intervals. Repeating the same analyses after cleaning demonstrated that much of this redundancy disappeared once post-window and duplicate variables had been removed. No pair of remaining numeric predictors exceeded the predefined correlation threshold of 0.80. Strong categorical associations remained between activity cluster and surgical specialisation, and between activity cluster and primary procedure category, and these relationships were carried forward for review during pre-modelling rather than being removed automatically.

Importantly, after exclusion of post-window variables, no single remaining feature showed a very strong univariate relationship with infection. This indicates that the final prediction problem is genuinely multivariate: infection risk must be estimated from the combined contribution of several relatively weak signals rather than from one dominant predictor.

Clinical text models

The three clinical-text methods were evaluated on a deliberately constructed shared evaluation set of 319 notes, containing 29 positive and 290 negative cases. This evaluation set was created from the full export of 35,925 surgical records, prior to the cohort restriction, by reserving

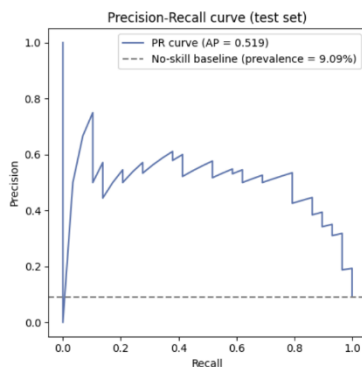
approximately 20% of the positive cases and sampling ten negative cases for each positive case. Its positive prevalence was therefore intentionally increased to 9.09%, compared with the much lower prevalence in the full surgical cohort. The purpose of this subset was not to estimate population-level clinical performance, but to enable a stable and consistent head-to-head comparison between the three text-processing methods under the same evaluation conditions. The NLP pipeline itself was subsequently applied across the full cleaned cohort used in the project.

Because the positive class was extremely small, reserving 29 positive cases for evaluation left only 112 positive cases available for training the text classifiers. This should be considered when interpreting the stability of the reported NLP results.

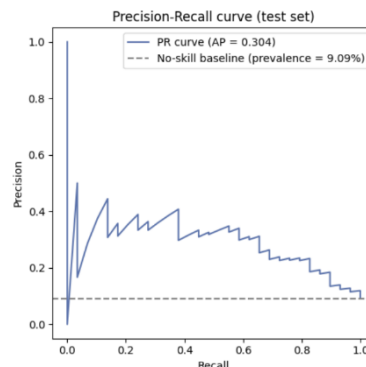
Table 4. Standalone performance of the three clinical-text methods on the shared evaluation split. Precision, recall and F1 are reported for the infection class at each condition's own operating point: a probability of 0.5 for the two supervised baselines and a self-reported risk score of 0.5 for the prompted model.

Text methods	ROC-AUC	PR-AUC (AP)	Precision	Recall	F1
TF-IDF + logistic regression	0.933	0.519	0.577	0.517	0.545
TF-IDF + XGBoost	0.821	0.304	0.286	0.069	0.111
Llama 3.3:70b, prompted (frozen inference)	0.701	0.183	0.169	0.345	0.227
No-skill baseline	0.500	0.091	-	-	-

TF-IDF + LOGISTIC REGRESSION



TF-IDF + XGBOOST



LLAMA

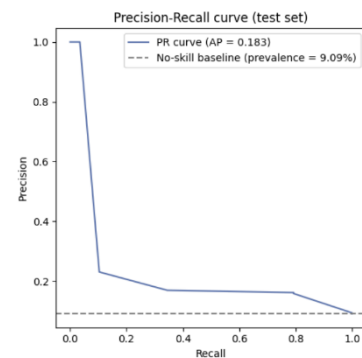


Figure 4. Precision-recall curves for the three clinical-text conditions on the shared evaluation split. All three curves lie above the no-skill baseline corresponding to the 9.09% positive prevalence of the comparison set. The TF-IDF and logistic-regression curve dominates across most of the recall range.

All three conditions exceed the no-skill average precision of 0.091 (Table 4, Figure 4), so Hebrew clinical narrative does carry infection signal, and the frozen inference completed for all 31,849 cohort cases with no parsing or call failures. Two further observations follow from these results. First, the simplest text model performed best despite the availability of a large contextual language model. One plausible explanation is that the TF-IDF model learned directly from the vocabulary and documentation patterns of this specific hospital corpus, whereas the prompted LLM was used in frozen inference mode and was not adapted to the local clinical language or documentation style. The relatively high level of templated and repeated wording observed during textual EDA may also favour a frequency-based representation.

Second, the difference between the two TF-IDF models shows that the choice of classifier matters even when the underlying text representation is held constant. Logistic regression achieved substantially higher average precision than XGBoost on the same sparse TF-IDF representation. With a small number of positive cases and a high-dimensional sparse matrix, the regularised linear classifier appears better matched to this particular representation.

The purpose of this stage was therefore not to establish the clinical performance of a standalone text classifier on the true surgical population, but to select the text representation to be carried forward into the multimodal pipeline. Based on this comparison, the TF-IDF logistic-regression representation was selected.

Structured tabular models

The four candidate algorithms were first evaluated under cross-validation on the training partition using a common reference configuration before full balancing selection and hyperparameter optimisation. This initial comparison (Table 5) was intended to provide a baseline view of the behaviour of logistic regression, random forest, XGBoost and CatBoost under the same training conditions.

Table 5. Reference cross-validated comparison of the four candidate algorithms with class weighting, before balancing selection and tuning. The Dummy row is the no-information reference: average precision close to the 1.19% prevalence of the prototype-aggregated training set, and ROC-AUC 0.5.

Model	CV PR-AUC (mean ± sd)	CV ROC-AUC	CV recall	CV precision
Logistic regression	0.315 ± 0.035	0.920	0.849	0.059

Model	CV PR-AUC (mean $\pm$ sd)	CV ROC-AUC	CV recall	CV precision
Random forest	0.282 $\pm$ 0.062	0.885	0.221	0.629
XGBoost	0.216 $\pm$ 0.049	0.895	0.303	0.312
CatBoost	0.209 $\pm$ 0.040	0.890	0.252	0.220
Dummy (prior)	0.012	0.500	0.000	0.000

Logistic regression achieved the highest mean cross-validated PR-AUC in this initial comparison, at 0.315, followed by random forest, XGBoost and CatBoost. The recall and precision values shown at the default 0.5 threshold differed substantially across algorithms, despite smaller differences in ranking metrics. This illustrates an important property of the task: threshold-dependent clinical behaviour cannot be inferred directly from ROC-AUC or PR-AUC, and the operating threshold therefore had to be selected separately from the algorithm itself.

The balancing-strategy screen then evaluated the prototype-aggregated training set as the baseline condition and compared three additional strategies: automatic class weighting, SMOTE and SMOTENC. Each strategy was evaluated using logistic regression and random forest under three-fold cross-validation, without hyperparameter tuning, so that the choice of balancing strategy could be made before the considerably more expensive full model search.

Table 6. Balancing-strategy screen on the training partition. Each strategy is averaged over two representative models, logistic regression and random forest, under three-fold cross-validation with no hyperparameter search; resampling is performed inside the fitting step, so only the training portion of each fold is augmented.

Strategy	Mean PR-AUC	Mean rank	Mean recall	Mean precision
SMOTE	0.300	1.0	0.566	0.289
Class weighting	0.296	2.0	0.515	0.358
Prototype aggregation only	0.284	3.0	0.227	0.618
SMOTENC	0.221	4.0	0.485	0.165

SMOTE achieved the highest mean PR-AUC and the best mean rank across the two representative algorithms and was therefore selected for the subsequent tuning stage. Class weighting produced a very similar average precision, whereas the prototype-only condition retained higher precision but substantially lower recall. SMOTENC performed worse than the other evaluated strategies.

The screen should not be interpreted as a comparison between balancing and no balancing: prototype aggregation had already been applied to the training partition before every condition in Table 6. Instead, the question was whether an additional balancing mechanism should be applied on top of that population-preserving reduction.

With the balancing strategy fixed, the four candidate algorithms were then tuned independently using RandomizedSearchCV (Table 7). Twenty hyperparameter configurations were evaluated for each model under five-fold stratified cross-validation, using average precision as the optimisation metric.

Table 7. Tuned cross-validated comparison under the locked SMOTE strategy. The final algorithm is selected from this table; the runtime column is reported because the near-tie between the two leading models was resolved in favour of the cheaper and more interpretable one.

Model	Tuned CV PR-AUC	Tuning runtime (s)
Logistic regression (locked)	0.329	11
CatBoost	0.325	1,547
XGBoost	0.289	31
Random forest	0.289	86

The locked model - logistic regression under SMOTE, with isotonic calibration and a decision threshold of 0.010 selected from calibrated out-of-fold training predictions - was then evaluated once on the frozen test partition of 9,555 surgical events containing 42 positive cases. Table 8 also reports all four tuned models at the default 0.5 cut-off, for transparency rather than as a selection step.

Table 8. One-time evaluation on the frozen test partition (9,555 surgical events, 42 positive, prevalence 0.440%). The upper rows show the four tuned models at the default cut-off; the final row is the locked model at the threshold selected from calibrated out-of-fold training predictions.

Model	Thr.	PR-AUC	ROC-AUC	Prec.	Recall	F2	TP	FP	FN	Alerts /1000
Dummy (prior)	0.50	0.004	0.500	0.000	0.000	0.000	0	0	42	0.0
Logistic regression	0.50	0.186	0.944	0.045	0.714	0.180	30	636	12	69.7
Random forest	0.50	0.259	0.947	0.283	0.357	0.339	15	38	27	5.5
XGBoost	0.50	0.163	0.914	0.234	0.262	0.256	11	36	31	4.9

Model	Thr.	PR-AUC	ROC-AUC	Prec.	Recall	F2	TP	FP	FN	Alerts /1000
CatBoost	0.50	0.226	0.935	0.231	0.357	0.322	15	50	27	6.8
Logistic regression - final, calibrated and tuned	0.010	0.183	0.943	0.047	0.881	0.193	37	755	5	82.9

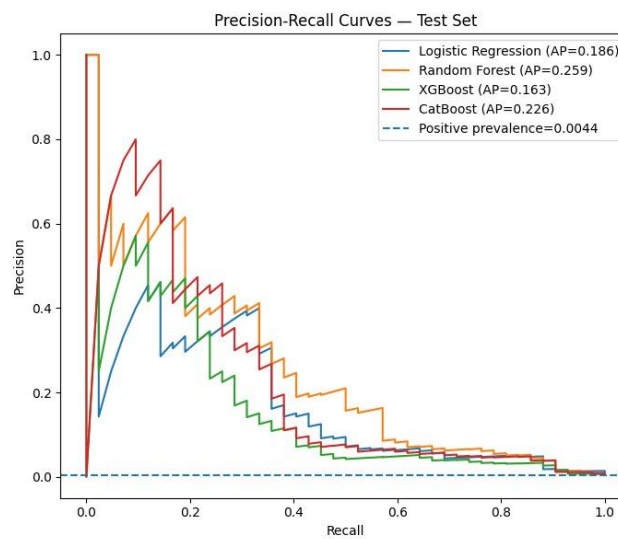


Figure 5. Precision-recall curves of the four tabular models on the frozen test partition. The curves illustrate model performance across decision thresholds under the highly imbalanced outcome distribution.

The final row is the result of the project: it identifies 37 of the 42 infections at a precision of 4.7%, which corresponds to a number needed to screen of 21.4 and 83 alerts per thousand operations - marginally outside the clinical criterion of 20 stated in the Evaluation, threshold tuning, calibration and explainability section, which the same threshold satisfied on the training data at 17.9. Balanced accuracy is 0.901 and the Brier score 0.0039. Figure 5 presents the precision-recall curves of the four tabular models on the same frozen partition, showing how each model trades precision against recall across the full range of decision thresholds. Three further properties of Table 8 deserve emphasis. First, discrimination is uniformly high and uninformative: all four models attain a ROC-AUC between 0.914 and 0.947 while their average precision ranges from 0.163 to 0.259, which is the arithmetic signature of a rare outcome and the reason average precision, not ROC-AUC, was the selection metric. Second, the threshold decision matters more than the choice of algorithm: the same logistic-regression model moves from 30 true positives and 636 false alerts at the default cut-off to 37 true positives and 755 false alerts at the calibrated clinical threshold, while the random forest at its default cut-off

achieves the highest average precision of any row and the second-worst recall. Third, calibration worked as intended. The Brier score of the selected model on the test partition improves from 0.0481 uncalibrated to 0.0039 after isotonic regression, a reduction of 92%, and now sits marginally below the 0.0044 of a predictor that simply returns the prevalence - which is the relevant comparison, since on a 0.44% outcome the prevalence itself is a very hard Brier score to beat.

The calibration method was selected using the training partition only. No calibration, sigmoid (Platt) scaling and isotonic regression were compared using Brier score, and isotonic regression achieved the best calibration performance and was therefore selected. Unlike Platt scaling, isotonic regression does not assume a fixed parametric relationship between the original model scores and calibrated probabilities, which provides greater flexibility when correcting miscalibration. Only the selected calibration method was then applied once to the frozen test partition.

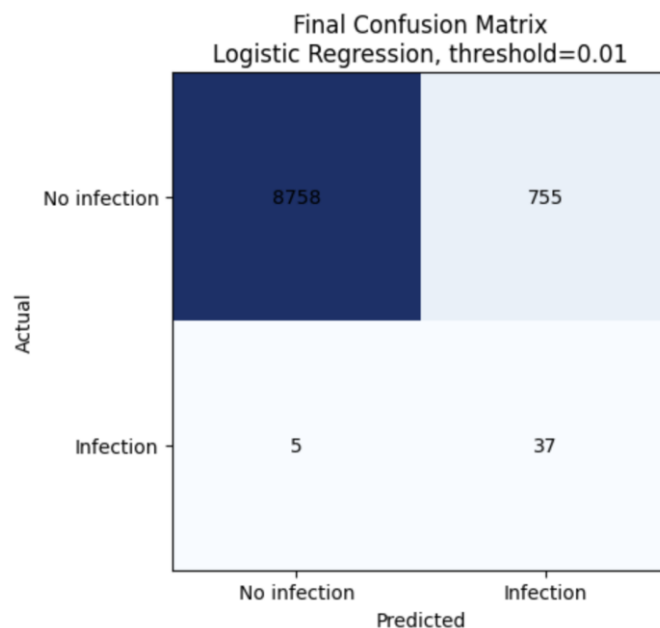


Figure 6. Confusion matrix of the final model on the frozen test partition. Logistic regression under SMOTE with isotonic calibration, at the locked decision threshold of 0.010.

The confusion matrix in Figure 6 cross-tabulates the true class against the predicted class at the locked threshold, so the diagonal holds the correctly classified cases and the off-diagonal the two kinds of error. Of the 42 infections in the partition the model identifies 37 and misses 5, while flagging 755 of the 9,513 non-infected operations. The asymmetry is the intended consequence of a threshold selected to maximise recall subject to a screening-workload ceiling, and it corresponds to 83 alerts per thousand operations.

Multimodal contribution, explainability and error analysis

The primary research question asks whether adding narrative-derived features to a structured model improves it. The comparison was executed: the locked algorithm was trained twice under an identical cross-validation framework, once on all twenty selected features and once on the nineteen that remain when the text-derived score is withheld (Table 9).

Table 9. Structured-only against structured plus the text-derived risk score. Both rows use the locked algorithm and the locked balancing strategy; the only difference is the presence of the single out-of-fold TF-IDF and logistic-regression score. One asymmetry should be noted when reading the table: the twenty features were ranked with the text-derived score present, so the structured-only row is the same feature set minus one column rather than an independently selected nineteen-feature set, which if anything favours the multimodal row.

Feature set	Features	CV PR-AUC (mean $\pm$ sd)	CV ROC-AUC	CV recall	Test PR-AUC	Test ROC-AUC
Structured + text-derived score	20	0.329 $\pm$ 0.035	0.921	0.869	0.186	0.944
Structured only	19	0.316 $\pm$ 0.035	0.916	0.859	0.190	0.949

Cross-validation showed a small but consistent gain in favour of the multimodal feature set - 0.012 in average precision, 0.005 in ROC-AUC and 0.010 in recall - while the frozen test partition showed a marginal difference in the opposite direction; with 42 positive test cases, neither difference is distinguishable from noise. The answer to the primary research question is therefore that on this cohort, with the narrative compressed into one scalar feature, no multimodal benefit could be demonstrated. The direction of the cross-validation result is favourable and consistent across folds, and the text-derived score does carry real signal - it is the fifth most influential of the twenty features by permutation importance and the sixth by mean absolute SHAP value - but the increment is an order of magnitude smaller than the differences reported in the English-language multimodal literature, and it does not survive into a test partition of this size. Three explanations are distinguishable in principle but not with these data. The narrative may simply add little beyond what the structured fields already record; or a single scalar may be too narrow a channel to carry what the text contains, given that the same TF-IDF representation reaches an average precision of 0.519 when it is allowed to classify directly. A third possibility is that the algorithms available under the constraints of this project were not the best suited to this task, so that a model family better matched to sparse, high-dimensional clinical data might extract more from the same features.

Explainability was produced in three layers, and they agree on the substance while differing in emphasis (Table 10). The L1 penalty retained by the tuned model set exactly half of the twenty coefficients to zero, so the effective model is a ten-feature linear score. Its largest terms are the

admission-screening fields classified as pre-operative in the Data cleaning and prevention of data leakage section, thoracic surgery as a specialisation, the general-surgery activity cluster, the presence of a urinary catheter, a raw primary procedure code, the text-derived score and age. Permutation importance on the test partition ranks the two detected-pathogen indicators an order of magnitude above everything else, whereas the SHAP attribution (Figure 7) ranks the activity cluster and the surgical specialisation first and the pathogen indicators near the bottom. The divergence is informative rather than contradictory: permutation importance measures how much average precision depends on a feature, and the pathogen indicators are sparse but decisive when present, while mean absolute SHAP measures how much a feature moves the score of a typical patient, which favours features that vary across the whole population.

Table 10. The three explainability layers for the final model, restricted to the features with a non-zero coefficient. Permutation importance is the mean drop in average precision over ten repetitions on the test partition; negative values mean the feature's observed arrangement is no better than a random one, which on 42 positive cases is within noise. Odds ratios are the exponentiated coefficients of the fitted logistic regression on standardised inputs. Mean absolute SHAP values are computed with a linear explainer over a 500-row sample of the test partition.

Feature	Permutation importance	Odds ratio	Mean SHAP
detected_pathogen - other organism	0.133	3.08	0.007
detected_pathogen - unknown	0.101	0.50	0.012
Primary procedure code (ICD-9)	0.009	0.89	0.085
Age	0.006	1.05	0.050
Text-derived risk score (nlp_tfidf_logreg_risk_score)	0.004	1.09	0.053
Surgeon specialisation - thoracic surgery	-0.015	2.32	0.338
Activity cluster - general surgery	-0.017	0.66	0.439
MRSA screening result - unknown	-0.005	0.62	0.187
Urinary catheter present	-0.021	1.28	0.146

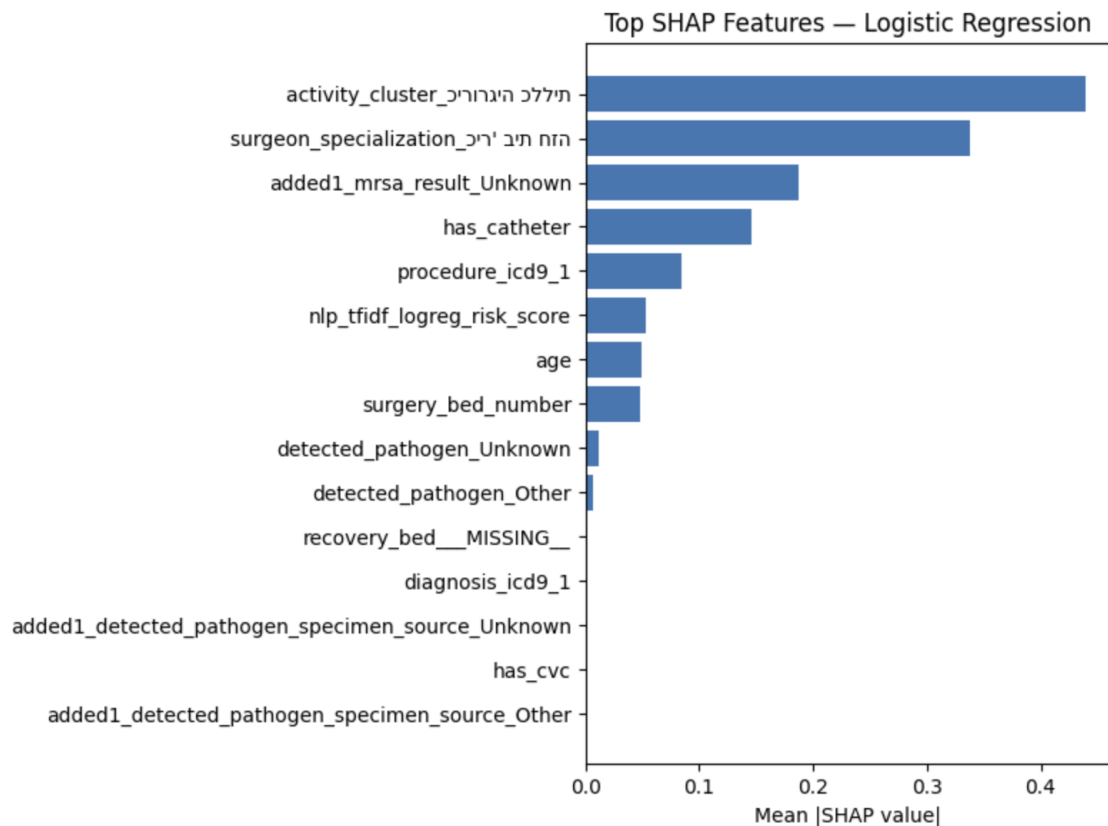


Figure 7. Mean absolute SHAP values of the final model. Computed with a linear explainer over a 500-row sample of the frozen test partition. Features whose coefficients the L1 penalty set to zero are omitted.

Clinically, the retained feature set is coherent. The admission-screening fields are pre-operative by construction, so a documented colonising organism at admission acting as a strong risk marker is clinically plausible rather than evidence of outcome leakage. The negative association observed for the “unknown” level of the same field may reflect differences in documentation intensity, since screening may be recorded more completely in patients who are followed more closely. Thoracic surgery as a high-risk specialisation, general surgery as a comparatively lower-risk cluster, the presence of a urinary catheter, and increasing age are all consistent with risk patterns identified in the literature review.

The error analysis of the final row of Table 8 is equally instructive. Five infections were missed at a threshold chosen to maximise recall, and all five were missed decisively rather than marginally: their calibrated probabilities lie between 0.00087 and 0.0019, an order of magnitude below the 0.010 threshold, so no small adjustment of the cut-off would have recovered them. The 755 false alerts cluster at a small number of repeated probability values, the highest at 0.614, which is a direct consequence of a ten-feature model in which most terms are binary: many patients share an identical feature pattern and therefore receive an identical score. Practically, these repeated scores create a limited number of risk strata that could potentially simplify review, but they also reduce the model’s ability to discriminate between patients within the same stratum.

Subgroup analysis on the test partition shows that the aggregate figures conceal substantial variation in performance (Table 11). The model performs best in the general-surgery cluster, where average precision reaches 0.424 at a precision of 28.6%, whereas performance is markedly weaker in thoracic surgery, where average precision is 0.029 and ROC-AUC is 0.568, close to chance, despite that cluster containing 15 of the 42 positive test cases and despite thoracic surgery carrying one of the largest coefficients in the model. In other words, thoracic specialisation increases the predicted risk level, but the model has limited ability to rank patients within that subgroup. Performance is also markedly weaker in the physiologically sicker group: average precision is 0.302 where the high-ASA indicator is zero and 0.113 where it is one. The spread between the best and worst activity cluster is larger than the spread between any two models in Table 8, which is the practical argument for stratification: these findings suggest that subgroup-specific or stratified modelling may be worth investigating in future work.

Table 11. Subgroup performance of the final model on the frozen test partition. Subgroups with fewer than twenty test events are omitted, as are those containing no positive case, for which average precision is undefined. All figures are at the single locked threshold of 0.010; no subgroup-specific threshold was fitted.

Subgroup	n	Positive	PR-AUC	ROC-AUC	Recall	Precision
Activity cluster - general surgery	6,690	14	0.424	0.995	0.857	0.286
Activity cluster - orthopaedics	2,112	13	0.277	0.821	0.769	0.085
Activity cluster - thoracic surgery	629	15	0.029	0.568	1.000	0.024
High ASA = 0	6,593	19	0.302	0.944	0.789	0.103
High ASA = 1	1,733	21	0.113	0.849	0.952	0.032
BMI - overweight	2,805	17	0.319	0.937	0.882	0.054
BMI - obesity	2,007	12	0.178	0.948	0.917	0.061
Department - operating theatre	9,475	40	0.199	0.945	0.875	0.048

Comparison with the project proposal and interim report

HIT requires that the results be compared explicitly with what was proposed. Table 12 does so item by item, and the deviations are then explained.

Table 12. Commitments in the project proposal and interim report against what was delivered. The table summarises the main planned objectives, the corresponding final implementation, and the status of each commitment.

Proposed / interim commitment	What was delivered	Status
Predict HAI risk during hospitalisation from structured EHR data	Narrowed to surgical-site infection in postsurgical patients, predicted at the end of surgery	Scope narrowed deliberately - the supplied data are surgical and the label is a wound-infection label
Full preprocessing pipeline with imputation, feature engineering and exploratory analysis	Delivered in full, with leakage control and a clinically classified missingness policy	Achieved
Three prediction checkpoints evaluated in parallel (pre-operative, 24 h, 48 h)	A single checkpoint at the end of surgery	Deviation - the exploratory analysis showed post-operative fields to be populated because suspicion already existed (see the Definition of the prediction target and prediction window section)
Compare logistic regression, random forest, XGBoost and CatBoost	All four tuned and evaluated; logistic regression locked from cross-validation	Achieved
Address imbalance with resampling and synthetic data (SMOTE, CTGAN)	Prototype aggregation on the training partition, then a screened choice of SMOTE; CTGAN not executed	Partial - CTGAN was not included in the final comparison because the modelling matrix had already been one-hot encoded, the package was also unavailable in the closed hospital environment.
Fine-tune a Hebrew Transformer (AlephBERT) on the clinical notes	Replaced by frozen large-language-model inference plus two TF-IDF baselines	Deviation - fine-tuning a Hebrew transformer was not feasible within the available computing environment; frozen LLM inference and TF-IDF baselines were used instead.
Test whether NLP features improve the structured model	Executed: structured-only against structured plus the text-derived score	Achieved but inconclusive - a consistent cross-validation gain, no test-set gain (Table 9)
Explainable AI across the architecture (SHAP and feature importance)	Coefficients and odds ratios, permutation importance, SHAP, error analysis and subgroup analysis	Achieved
A validated predictive model with potential clinical utility	A leakage-controlled pipeline and a locked model achieving 88.1% recall, 4.7% precision and NNS 21.4 on the frozen test set	Partially achieved - the predictive signal is meaningful, but prospective clinical validation was not performed.

Most core objectives were achieved, including development of the full preprocessing pipeline, evaluation of the planned model families, implementation of imbalance-handling methods, integration of clinical text, and explainability analysis. The principal deviations concerned the



narrowing of the prediction target, the use of a single end-of-surgery prediction point instead of three parallel checkpoints, the replacement of planned Hebrew-transformer fine-tuning with TF-IDF and frozen LLM inference, and the non-use of CTGAN in the final balancing comparison. These deviations arose from characteristics of the supplied data, temporal-leakage findings, and constraints of the hospital computing environment.

The most consequential deviation was the abandonment of the three-checkpoint design. Exploratory and temporal analysis showed that several postoperative laboratory values, vital signs and clinical notes were recorded only after clinical suspicion had already emerged. Including these variables would therefore have risked allowing the model to reproduce an existing suspicion rather than predict infection prospectively. For this reason, the final prediction point was fixed at the end of surgery and the retained feature set was limited primarily to information available before or during the procedure.

Discussion

Interpretation of the results against the literature

The discrimination achieved by the four tabular models was high, with ROC-AUC values ranging from 0.914 to 0.947. At first glance, these values are comparable with the upper range reported in the surgical-site-infection prediction literature. However, this comparison should be interpreted carefully, because the present study differs from many published studies in outcome prevalence, prediction timing and the information available to the model.

A first important consideration is the extreme class imbalance in the current cohort. With an infection prevalence of only 0.44%, the negative class is very large, and a model may achieve a high ROC-AUC while still generating many false-positive alerts. This can be seen directly in Table 8. The final model achieved a ROC-AUC of 0.943, but its average precision was 0.183 and its precision at the selected clinical operating point was 4.7%. A published ROC-AUC of 0.944 in an intensive-care cohort with a prevalence of several per cent [16] is therefore not directly comparable with a ROC-AUC of 0.943 in the present cohort. In this setting, operational measures such as precision, alerts per 1,000 procedures and the number needed to screen provide a more direct indication of the burden that an infection-prevention team would face. The final model required approximately 21.4 flagged operations for each infection identified. The clinical criterion was satisfied on the training data and was missed only marginally on the frozen test partition, which provides a realistic indication of the model's current operating performance.

A second consideration is the difference in the feature set available to the model. Many published models that report similar discrimination use postoperative vital signs, repeated laboratory measurements and, in some cases, postoperative clinical narrative. These variables were deliberately excluded from the present model because the temporal analysis showed that, in this dataset, many of them were recorded only after clinical suspicion of infection had already emerged. The comparison with the literature is therefore not between models operating at exactly the same clinical time point. The model developed here makes its prediction at the end of surgery, using predominantly preoperative and intraoperative information, whereas some published models make predictions later in the postoperative course. Viewed in this context, identifying approximately 88% of infections using information available by the end of surgery is encouraging.

The findings from the clinical-text analysis also differed from what might have been expected from the literature. Studies reporting strong multimodal gains often use contextual language models on English-language clinical corpora. In the present study, however, the simpler TF-IDF representation combined with logistic regression performed better than the prompted large language model, reaching an average precision of 0.519 compared with 0.183 for the Llama model on the same evaluation set. Several factors may help explain this difference. The clinical notes were written in Hebrew, no domain-specific adaptation was performed for the frozen

language model, and the documentation contained a substantial amount of repeated and templated wording. Between 13% and 29% of notes were exact duplicates of another note, a pattern that may favour a frequency-based representation such as TF-IDF.

The multimodal result itself should be interpreted as unresolved rather than negative. Adding the text-derived score produced a small increase in cross-validated average precision, but this improvement was not reproduced on the frozen test partition. The test set contained only 42 positive cases, which limits the stability of small differences between models. In addition, the narrative information entered the tabular model as a single scalar score, whereas studies reporting larger multimodal gains often retain a richer text representation. The fact that the TF-IDF representation achieved an average precision of 0.519 as a standalone text classifier suggests that useful information was present in the notes, but that compressing this information into a single score may have limited its contribution to the final multimodal model.

The results also support the decision to fix the prediction point at the end of surgery. This choice reduced the amount of postoperative information available to the model, but it also reduced the risk of temporal leakage. The temporal analysis showed that several potentially informative postoperative variables were recorded only after a clinician had already developed a suspicion of infection. Including them could therefore have allowed the model to reproduce an existing clinical suspicion rather than predict risk prospectively.

Finally, the final results provide support for the leakage-control strategy used throughout the project. Variables known to occur after the prediction point were excluded, time-dependent variables were checked against the defined prediction window, admission-screening fields were reviewed clinically, and the text-derived score was generated using an out-of-fold procedure. Explainability analysis did not identify an obvious major residual leakage feature among the final influential predictors. This does not prove that leakage is completely absent, but it provides additional support that the final model relies primarily on information that would have been available at the intended prediction point.

Limitations

Statistical power. Forty-two positive cases in the frozen test partition provide a limited basis for estimating model performance. A shift of only three cases changes recall by more than seven percentage points, illustrating how sensitive the reported metrics are to a small number of observations. The results in the Project Results chapter should therefore be interpreted as estimates of overall performance rather than as highly precise values. The same limitation also makes the multimodal comparison in Table 9 difficult to interpret conclusively in either direction.

The feature set was reduced to twenty. The pre-modelling stage produced 266 columns, while the modelling stage retained the twenty highest-ranked features according to the combined mutual-information and L1 ranking. This reduction was reasonable given the small number of

positive training cases and also made the final model easier to interpret. Reducing the dimensionality was also intended to limit the risk of overfitting in a setting where the number of candidate predictors was large relative to the number of positive cases. Retaining a compact feature set therefore provided a practical balance between predictive information, model stability and clinical interpretability.

A further limitation concerns the evaluation of the clinical-text models. Approximately 20% of the positive infection cases were reserved for the shared NLP evaluation set, leaving only 112 positive cases available for training. Given the rarity of the outcome, this reduced the amount of positive-class information available to the text classifiers and may have limited their ability to learn stable infection-related patterns. At the same time, retaining an independent positive subset was necessary to enable a consistent comparison between the competing NLP methods. Consequently, the differences observed between the text models should be interpreted cautiously, particularly given that the evaluation set itself contained only 29 positive cases.

The text-derived feature represents only one implementation of multimodal integration. The score passed to the tabular model used vectoriser settings that were not identical to the configuration evaluated in the standalone text-model comparison, so the two pipelines were related but not exactly the same. In addition, the split used to compare the three text methods was drawn from the full export and overlapped with the later modelling partitions. Therefore, although the text score itself was generated out-of-fold with respect to the locked modelling split, the earlier choice between text methods was not completely blind to cases that later appeared in the frozen test partition.

The admission-screening classification rests on a clinical convention rather than a timestamp. The detected-pathogen, MRSA and specimen-source fields were retained as pre-operative. These fields became influential predictors in the final model. Unlike several device-related variables, however, they could not be verified against a patient-level timestamp because no corresponding dates were available in the export. Their classification is clinically justified and was made explicitly, but it remains a clinical judgement rather than a direct temporal verification and should therefore be confirmed against the source system before any prospective implementation.

The cohort restriction may reflect a data-quality artefact. The Final Study Cohort excludes four activity clusters in which no infections were observed. These clusters may genuinely represent very low-risk procedures, but zero recorded infections could also reflect differences in documentation, coding or follow-up outside the hospital. The restriction improved the definition of the modelling population, but it also narrowed the population to which the final model can be applied. Any future clinical use of the model should therefore take into account that it was trained and evaluated on this defined set of surgical activity clusters and may require additional validation before being applied to surgical populations that were excluded from the study cohort.

Label provenance remains a limitation, although the low observed infection prevalence should be interpreted in the context of the study setting. As established in clinical consultation with the hospital, Assuta Ramat HaHayal is a private medical center in which surgical candidates are selected in part to reduce the likelihood of complications, and the resulting case mix may therefore be lower-risk than that of broader surgical populations reported in the literature. The prevalence of approximately 0.44% should consequently not be interpreted as evidence of under-ascertainment on its own. The outcome was nevertheless derived from two diagnosis-related fields that were not populated consistently. In addition, complete follow-up after discharge was not available for every patient, so infections diagnosed or treated outside the hospital after discharge may not have been captured in the dataset. These limitations in documentation and follow-up mean that some postoperative infections may still be missing from the recorded outcome, and the reported performance measures should be interpreted in light of the available clinical documentation.

Single-center setting and lack of external validation. All data were obtained from a single private medical center. Its patient population, case mix, clinical pathways and documentation practices may differ from those of other hospitals, particularly public institutions. No external validation was performed, so the results cannot yet be assumed to generalise beyond the study setting.

The subgroup analysis in Table 11 showed substantial variation in model performance across clinical groups, with stronger performance in general surgery and considerably weaker performance in thoracic surgery. Despite this heterogeneity, a single pooled model and a single operating threshold were used for the full cohort. Future work should therefore investigate whether subgroup-specific models or thresholds could provide more stable performance.

Success and failure per objective

Table 13. Assessment of the five project objectives defined in the original project proposal against the final evidence produced.

Objective	Outcome	Evidence
1. Develop a data-driven machine-learning model to predict infection risk during hospitalization	Achieved within the final postoperative SSI scope	A complete leakage-controlled modelling pipeline was developed and evaluated on a frozen test partition of 9,555 surgical events. The final calibrated logistic-regression model achieved ROC-AUC 0.943, PR-AUC 0.183 and 88.1% recall. The original HAI-wide scope was narrowed to postoperative surgical-site infection according to the outcome information available in the supplied dataset.

Objective	Outcome	Evidence
2. Enable early identification of patients at elevated risk to support timely preventive intervention	Partially achieved	The final model identifies risk at the end of surgery, before postoperative infection is clinically documented, and detected 37 of 42 infections in the frozen test set (88.1% recall). However, the model was not prospectively deployed, and its effect on clinical intervention or patient outcomes was therefore not evaluated.
3. Identify key clinical and demographic variables associated with infection risk	Achieved	Feature-selection and explainability analyses identified clinically relevant predictors including age, surgical specialisation, urinary-catheter status and admission-screening variables. Their associations were examined using model coefficients, odds ratios, permutation importance and SHAP analysis.
4. Evaluate the contribution of these variables to predictive performance	Achieved	Variable contribution was assessed through mutual-information and L1-based feature selection, model coefficients and odds ratios, permutation importance, SHAP values and subgroup analyses. The contribution of clinical text was also tested by comparing the structured-only model with the structured model including the text-derived score.
5. Compare different predictive models and select the most accurate and reliable approach	Achieved	Logistic regression, Random Forest, XGBoost and CatBoost were tuned and compared using stratified cross-validation, with PR-AUC as the primary selection metric. Logistic regression achieved the highest cross-validated average precision under the selected balancing strategy and was therefore locked as the final algorithm before evaluation on the frozen test set.

As Table 13 shows, each of the five objectives defined in the project proposal is assessed individually against that evidence, and marked as achieved, achieved within the final postoperative SSI scope, or partially achieved. The objectives are evaluated according to the final project scope and the results obtained on the locked modelling and test partitions.

Conclusions

Five conclusions follow from this work.

1. Hebrew clinical narrative carries genuine infection signal, and the simplest representation extracted the most of it. All three text conditions beat the no-skill baseline, and TF-IDF with logistic regression led on every metric, reaching an average precision of 0.519 against 0.183 for a prompted seventy-billion-parameter model on the same notes. For this dataset, TF-IDF with logistic regression was the strongest of the text methods that could be evaluated within the available environment, showing that a simple and interpretable representation can be highly effective. However, this does not establish that frequency-based modelling is optimal for Hebrew clinical text in general, and comparison with a domain-adapted Hebrew transformer remains an important direction for future work.
2. At 0.44% prevalence, high discrimination does not necessarily translate into high precision. A ROC-AUC above 0.94 coexists here with an average precision of 0.18 and a number needed to screen of 21. For an infection-prevention setting, ROC-AUC or accuracy alone therefore provide an incomplete picture; precision, alert burden and number needed to screen are also needed to understand the practical implications of the model. The threshold decision, rather than the choice of algorithm alone, is where the clinically meaningful trade-off between detecting infections and generating false alerts is ultimately made.
3. The prediction point is a substantive clinical decision, not merely a technical parameter. The empirical finding that postoperative fields were often populated after clinical suspicion had already emerged forced the prediction window back to the end of surgery and invalidated two of the three checkpoints originally proposed. The cost was the loss of many variables commonly used in the literature; the benefit is that the final prediction represents an early warning rather than a restatement of an existing clinical suspicion.
4. A single scalar may be too limited a representation for multimodal integration. The text pipeline achieved an average precision of 0.519 on its own but contributed only 0.012 in cross-validated average precision when compressed into one feature, and this gain was not reproduced on the frozen test partition containing 42 positive cases. This does not indicate that the narrative is uninformative; rather, it suggests that the way the text was integrated into the tabular model may have limited its contribution.
5. Leakage control has to be designed into the pipeline, and the final workflow combined four mechanisms: an unconditional drop list, a per-row empirical timing gate, explicit clinical classification of ambiguous microbiological fields, and out-of-fold generation of the text-derived score against a split created before that score existed. Together, these steps provide support that the influential features in the final model were available at

the intended prediction point. Explainability was used as an additional check for suspicious predictors, rather than as evidence that leakage was completely absent.

What the project delivers, therefore, is not a deployable model. It is a leakage-controlled and fully documented pipeline, a defensible feature set, an empirically justified prediction point, a validated Hebrew text component, an executed multimodal comparison, an evaluation framework adapted to the very low outcome prevalence, and a clear account of the additional validation and development required before clinical implementation.

Further research and development

In the short term, one of the main directions for further research is to broaden the way textual information is integrated into the multimodal model. Instead of passing a single scalar score to the tabular model, future work could examine a broader TF-IDF representation, several separate scores derived from different text fields, or additional structured features extracted from the clinical narrative. The comparison with the structured-only model should then be repeated under this richer integration strategy.

In parallel, the choice of twenty retained features should be examined empirically by comparing the current model with alternative feature-set sizes, including intermediate subsets and the wider feature matrix, while performing feature ranking within the cross-validation folds. A more extensive optimisation of hyperparameters, calibration methods, decision thresholds and complementary evaluation metrics should also be performed. The aim would be to identify the combination that preserves high sensitivity to infection while reducing false-positive alerts as much as possible, thereby improving the clinical operating point of the model. Subgroup-specific thresholds or models should also be explored, with particular attention to thoracic surgery, where performance was substantially weaker despite a relatively large number of positive cases.

Another short-term direction is to evaluate a Hebrew Transformer model adapted to the clinical domain. The current comparison does not establish that TF-IDF is generally superior, because the language model used in this project operated in frozen-inference mode and was not fine-tuned on the local clinical text. If GPU access becomes available within the secure environment, TF-IDF should be compared directly with a Hebrew BERT-based model or a similar Transformer fine-tuned on relevant clinical text. This would help determine whether the observed advantage of TF-IDF reflects the characteristics of the current corpus or the limitations of the language model that could be evaluated in this project.

A further short-term direction is to revisit the treatment of missing values in the boosting models. Since models such as XGBoost and CatBoost have native mechanisms for handling missing values, they could be evaluated on a processed version of the dataset in which median imputation is not performed. Their performance could then be compared with the current

imputed pipeline to determine whether preserving the original missingness patterns improves prediction.

CTGAN should also be investigated more thoroughly as an alternative approach to class imbalance. In the current project it was not included in the final comparison because of the combination of constraints in Assuta's closed computing environment and the fact that the modelling matrix had already undergone one-hot encoding. Future work could examine a pipeline in which CTGAN is applied before one-hot encoding where appropriate, as well as controlled combinations of CTGAN with other balancing methods such as SMOTE.

In the longer term, the model could be extended from a single fixed prediction point to a dynamic prediction framework in which risk is updated as new postoperative information becomes available. Such an approach would require more precise temporal data and a substantially larger number of positive cases, while maintaining strict temporal controls to ensure that the model does not learn from information recorded only after clinical suspicion has already emerged.

After further refinement and validation of the model, the next stage would be to evaluate it prospectively and actively in surgical departments. The model could be applied to newly admitted surgical patients in real time, with its alerts recorded and its performance monitored against subsequent clinical outcomes.

At the stage of clinical implementation, the model's risk score could be incorporated as an additional feature in an existing hospital dashboard. Patients identified as being at high risk could be highlighted for the relevant clinical team, supporting more focused monitoring of cases that may require additional attention. At this stage, evaluation should address not only predictive performance but also how well the alerts integrate into the clinical workflow and whether they contribute to earlier identification of high-risk patients.

Summary of project activities and final system specification

The project was conducted in two phases, separated by the point at which the dataset became available. In the first phase, while the data were not yet accessible, the methodology was defined based on the literature review and the supplied data dictionary: model families were selected, the evaluation strategy was established, and the expected outcomes were specified in advance, as documented in the interim report and in the methodology sections. In the second phase, once the data became available, the original plan was adapted to the characteristics observed in practice. The prediction window was defined empirically, the target label was constructed from two diagnosis fields that were populated inconsistently, the advanced NLP component was adapted from fine-tuning to frozen inference, the text representation carried forward was selected based on measured performance rather than architectural complexity, and the leakage-control mechanisms were built around a locked train/test split created before any learned features were generated.



The delivered system is an analytical pipeline consisting of eight Python notebooks, together with a manually reviewed semantic type mapping, a clinically approved medication-family vocabulary, a predefined activity-cluster exclusion list used for cohort definition, a locked train/test split file, a case-level text-derived risk score, cached language-model outputs for all 31,849 cases, trained model artefacts, and audit files produced throughout the different stages of the workflow.

The final model specification is a Logistic Regression classifier based on 20 selected features, trained on a prototype-aggregated training partition containing 8,288 surgical events, with SMOTE applied during model fitting. The model was calibrated using isotonic regression and operated at a decision threshold of 0.010, selected according to a clinical alert-burden criterion. On a frozen test partition of 9,555 surgical events containing 42 infections, the model identified 37 of them, corresponding to a recall of 88.1%, with a precision of 4.7%, approximately 83 alerts per 1,000 operations, a number needed to screen of 21.4, a PR-AUC of 0.183, a ROC-AUC of 0.943, and a Brier score of 0.0039.

Overall, the final testing indicates that the model captures meaningful predictive signal, that the leakage-control mechanisms operated as intended, and that the selected operating point came close to the predefined clinical workload criterion. At the same time, several clear directions for further improvement remain, including richer integration of textual information, more extensive optimisation of hyperparameters, calibration methods and decision thresholds, and further evaluation of alternative preprocessing and balancing strategies. Accordingly, the final project deliverable is a complete, documented and extensible analytical framework that is ready for further refinement, validation and progression toward prospective clinical evaluation.



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